



**School of Computer Science and Electronic  
Engineering**

**MSc Data Science**

**Academic Year 2024-2025**

**Classical machine learning versus deep learning for data-  
scarce Cropland Mapping: A Case Study of Iran**

A project report submitted by: Mohammad Shayaan Aslam Khan 6941885

A project supervised by: Dr. Mahdi Mashhadi

A report submitted in partial fulfilment of the requirement for the degree of Master of  
Science

University of Surrey  
School of Computer Science and Electronic Engineering  
Guildford, Surrey GU2 7XH  
United Kingdom.  
Tel: +44 (0)1483 300800

## ABSTRACT

Food security and monitoring agricultural policy and environmental management are all supported by accurate and current farmland extent maps; nevertheless, current global land-cover products are static, inconsistently defined and often contradict each other. In order to overcome these constraints, this dissertation uses multi-sensor satellite imagery (Sentinel-1, Sentinel-2 and Landsat-8) to develop and assess machine learning and deep learning techniques for annual cropland extent mapping at 10-metre resolution. Iran is chosen as a detailed case study region.

Building on the ITU/FAO/UNODC GeoAI Challenge for cropland mapping, the work focuses implementation on Iran alone to allow for a more rigorous treatment of temporal extendibility within an MSc timeframe. It does this by adopting the challenge's dataset, evaluation protocol and limited training budget. In order to test the hypothesis that deep learning provides a particular advantage in discriminating spectrally similar cropland classes rather than necessarily improving overall accuracy, three tuned gradient-boosting and ensemble baselines (CatBoost, XGBoost and Random Forest), consistent with the strongest publicly documented solutions to the original challenge, were compared against a temporal convolutional deep learning model. The disproportionate emphasis of crop-mapping research on data-rich contexts compared to the semi-arid, data-scarce location taken into consideration here is a gap in the reviewed literature that this work fills.

In contrast, both during the training season (Cohen's Kappa 0.94 for CatBoost against 0.92 for the temporal CNN) and, more decisively, when compared to a second, unseen year of imagery at the same locations (0.828 against 0.796), classical gradient boosting matched and sometimes slightly outperformed the deep learning model's performance. This difference widened rather than narrowed under this more practically demanding test. Although it did not reach conventional statistical significance due to the small sample available, this directional advantage held consistently across five-fold and higher repeated cross-validation as well as a sensor-ablation study. This is an expected result of the training sample constraint rather than a flaw in the evaluation.

Instead of confusing cropland with obviously distinct land cover, both model families made the vast majority of their mistakes on surfaces that were actually cropland-like, as determined by an independent global land-cover product used in place of the crop-type labels the dataset does not offer. The dissertation concludes with a discussion of how the pipeline would need to be modified for Sudan and Afghanistan, two areas that are not specifically examined here. It concludes that each region presents a significantly different agricultural and temporal setting, necessitating direct revalidation rather than assuming transfer of these Iran-specific results. This paper provides a methodologically sound negative for deep learning data points to a corpus of literature that typically reports the opposite.

## **HIGHLIGHTS**

- First documented deep learning baseline for the 2023 GeoAI Cropland Mapping Challenge dataset
- CatBoost/RandomForest/XGBoost outperform a tuned TempCNN under a 500-sample training cap.
- The classical advantage widens, not narrows, under cross-season temporal generalisation testing.
- A genuine evaluation leakage bug was found, diagnosed, and corrected mid-project.
- WorldCover substitute analysis shows errors cluster within cropland-like surfaces only.
- Critical scalability discussion grounded in Sudan and Afghanistan's actual agricultural systems

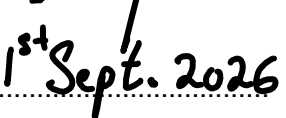
## ACKNOWLEDGEMENTS

I would like to thank my supervisor, Dr. Mahdi Mashhadi, for providing the topic and initial direction for this dissertation.

I would also like to thank my family and friends for their support and encouragement throughout my MSc and during the completion of this project.

Finally, I am grateful for the opportunity to undertake this research independently. Completing the project required me to take responsibility for the research process, including the literature review, data collection and preparation, implementation, evaluation, troubleshooting, and critical analysis of the results. This experience has been a valuable part of my academic and personal development.

I certify that the work presented in the dissertation is my own unless referenced

Signature.....  
Date.....

TOTAL NUMBER OF WORDS: 13082

**TABLE OF CONTENTS**

|                                                                                                     |    |
|-----------------------------------------------------------------------------------------------------|----|
| Table of Contents .....                                                                             | 5  |
| List of Tables .....                                                                                | 7  |
| CHAPTER 1: INTRODUCTION .....                                                                       | 8  |
| 1.1 Background .....                                                                                | 8  |
| 1.2 Research aim and objectives .....                                                               | 9  |
| 1.3 Research approach.....                                                                          | 10 |
| 1.4 Dissertation outline .....                                                                      | 10 |
| CHAPTER 2: LITERATURE REVIEW .....                                                                  | 11 |
| 2.1 Introduction.....                                                                               | 11 |
| 2.2 Existing Global Cropland Products and Their Limitations .....                                   | 11 |
| 2.3 Deep Learning in Remote Sensing: The Broader Landscape.....                                     | 11 |
| 2.4 Classical Machine Learning Approaches to Cropland Classification<br>.....                       | 11 |
| 2.5 Deep Learning Architectures for Crop and Cropland Mapping ....                                  | 12 |
| 2.6 Multi-Sensor Data Fusion.....                                                                   | 14 |
| 2.7 Learning from Limited Labelled Data .....                                                       | 14 |
| 2.8 Temporal Generalisation and Cross-Year Transferability .....                                    | 15 |
| 2.9 Deep Learning versus Classical Machine Learning: Synthesis of<br>the Comparative Evidence ..... | 15 |
| 2.10 Gap Analysis and Positioning of the Current Study .....                                        | 16 |
| 2.11 Summary .....                                                                                  | 16 |
| CHAPTER 3: RESEARCH APPROACH.....                                                                   | 17 |
| 3.1 Introduction .....                                                                              | 17 |
| 3.2 Choice of Methodology .....                                                                     | 17 |
| 3.3 CRISP-DM Applied to This Project.....                                                           | 17 |
| 3.4 Data Access and Development Environment .....                                                   | 18 |
| 3.5 Ethical Considerations .....                                                                    | 19 |
| 3.6 Summary .....                                                                                   | 19 |
| CHAPTER 4: DATA ANALYSIS .....                                                                      | 20 |
| 4.1 Introduction .....                                                                              | 20 |
| 4.2 Data Understanding .....                                                                        | 20 |
| 4.2.1 The GeoAI Challenge dataset .....                                                             | 20 |
| 4.2.2 Regional filtering.....                                                                       | 20 |
| 4.2.3 A data-quality issue in the provided starter material ....                                    | 20 |
| 4.3 Data Preparation.....                                                                           | 21 |
| 4.3.1 Sensors and reference period .....                                                            | 21 |
| 4.3.2 Bands, indices, and cloud masking.....                                                        | 21 |
| 4.3.3 Missing data .....                                                                            | 22 |
| 4.3.4 Data quality checks .....                                                                     | 22 |
| 4.4 Modelling .....                                                                                 | 22 |
| 4.4.1 Classical baselines.....                                                                      | 22 |
| 4.4.2 Deep learning model: architecture .....                                                       | 23 |
| 4.4.3 Deep learning model: Iteration 1.....                                                         | 23 |
| 4.4.4 Deep learning model: Iteration 2.....                                                         | 24 |
| 4.5 Evaluation .....                                                                                | 25 |
| 4.5.1 Comparative classification performance .....                                                  | 25 |
| 4.5.2 Feature importance .....                                                                      | 26 |
| 4.5.3 Sensor ablation .....                                                                         | 27 |
| 4.5.4 Discriminating spectrally similar surfaces .....                                              | 27 |

**CLASSICAL MACHINE LEARNING VERSUS DEEP LEARNING FOR DATA-SCARCE CROPLAND  
MAPPING: A CASE STUDY OF IRAN**

|                                                                               |    |
|-------------------------------------------------------------------------------|----|
| 4.5.5 Temporal extendibility.....                                             | 28 |
| 4.6 Summary .....                                                             | 29 |
| CHAPTER 5: DISCUSSION .....                                                   | 30 |
| 5.1 Introduction.....                                                         | 30 |
| 5.2 Comparison with Existing Research.....                                    | 30 |
| 5.3 Critical Evaluation of the Results.....                                   | 30 |
| 5.3.1 The methodological correction and what it demonstrates<br>.....         | 31 |
| 5.3.2 How much confidence the evidence supports.....                          | 31 |
| 5.4 Critical Evaluation of the Objectives.....                                | 31 |
| 5.5 Limitations .....                                                         | 32 |
| 5.6 Scalability to Sudan and Afghanistan.....                                 | 33 |
| 5.6.1 What transfers directly.....                                            | 33 |
| 5.6.2 Sudan: a closer fit in period, a different agricultural<br>system ..... | 33 |
| 5.6.3 Afghanistan: a genuinely different temporal design<br>problem .....     | 34 |
| 5.6.4 Summary.....                                                            | 34 |
| 5.7 Summary .....                                                             | 34 |
| 6. Conclusion .....                                                           | 35 |
| 6.1 Summary of the Dissertation.....                                          | 35 |
| 6.2 Research Contributions .....                                              | 35 |
| 6.3 Limitations and Future Research and Development.....                      | 36 |
| 6.4 Personal Reflections .....                                                | 37 |
| REFERENCES.....                                                               | 39 |
| APPENDIX A: ETHICAL APPROVAL.....                                             | 42 |
| APPENDIX B: OTHER APPENDICES.....                                             | 43 |

## LIST OF TABLES

|                                                                                             |    |
|---------------------------------------------------------------------------------------------|----|
| Table 4.1: Cross-validated Kappa, five-fold spatially blocked CV .....                      | 25 |
| Table 4.2: Repeated-CV Kappa (15 GroupShuffleSplit splits).....                             | 26 |
| Table 4.3: Pooled out-of-fold confusion matrices and accuracy .....                         | 26 |
| Table 4.4: Sensor ablation (Kappa, five-fold spatial CV).....                               | 27 |
| Table 4.5: Out-of-fold errors by WorldCover land-cover class .....                          | 28 |
| Table 4.6: Cross-season Kappa (trained Jul2019-Jun2020, evaluated<br>Jul2020-Jun2021) ..... | 28 |

## CHAPTER 1: INTRODUCTION

### 1.1 Background

Cropland extent maps, which are spatial records of the areas where crops are grown, are a fundamental component of many applications, such as forecasting food production; tracking progress toward the Sustainable Development Goals of the United Nations, especially SDG 2 (Zero Hunger) and SDG 15 (Life on Land); and monitoring agricultural expansion or abandonment. Traditionally, the creation of these maps has relied on national agricultural surveys and censuses, which are expensive, rare, and, in many cases, either unavailable or incomplete.

Cropland maps can now be created using publicly accessible imagery rather than just ground surveys thanks to developments in satellite Earth observation, particularly the European Space Agency's Copernicus Sentinel mission and the ongoing Landsat program, as well as cloud-based processing tools like Google Earth Engine. There are currently several global farmland products, such as GlobalLand30, GFSAD30, and WorldCover. However, these products have three recurring flaws that restrict their practical utility: they are not updated annually, making it impossible to track changes from year to year; they use inconsistent definitions of what constitutes cropland, none of which fully align with the FAO's own classification standards. They also often disagree when their outputs are directly compared over the same area (FAO, 2023).

The International Telecommunication Union, in collaboration with the Food and Agriculture Organization of the United Nations and the UN office on Drugs and Crime, launched the GeoAI challenge for cropland mapping in 2023 as part of its AI for Good programme in response to these constraints. Participants in the challenge were required to create annual farmland extent maps for Afghanistan, Iran and Sudan with a resolution of 10 metres, which is significantly finer than the standard 30 metres. Crucially, the challenge limited the number of labelled training samples that participants could access to 1,000 points per region, purposefully simulating the data-scarce circumstances that many under-resourced regions experience, where there are simply no large, high-quality labelled datasets (Zindi, 2023).

The dataset and constraints of the GeoAI Challenge serve as the foundation for this dissertation's structured academic investigation into the performance of machine learning and deep learning techniques under precisely these conditions. Specifically, it examines whether deep learning's well-established advantages in the crop-mapping literature translate into a significant advantage when the available training data is this limited. Iran is chosen as an in-depth case study out of the three original regions of the challenge because it provides the closest directly comparable published benchmark (a 2025 arid-region comparison of classical and deep learning methods (Mohammadi Mobarakeh, Pourmanafi and Ahmadi, 2025), discussed in Section 2.9). This enables the results of this dissertation to be evaluated against an existing, directly relevant body of evidence rather than just the leaderboard of the original challenge. Instead of being fully implemented, Sudan and Afghanistan are addressed in a critical discussion (Chapter 5), which reflects a purposeful scope decision to prioritise depth over breadth across all three regions, especially on the challenge's relatively under-explored temporal extendibility objective.

The vast majority of frequently cited research and benchmark datasets, like EuroCrops, are derived from data-rich, thoroughly examined agricultural systems in Europe or North America, despite a sizable and expanding body of literature on machine learning and

deep learning for crop and farmland mapping. Few studies look at performance in the more difficult semi-arid or dry climates found in Afghanistan, Iran, and Sudan, where cloud-free optical vision is not always guaranteed and field boundaries are frequently tiny, fragmented and irregular. A second, more focused gap stems directly from the public record of the original 2023 challenge itself: all publicly documented top-placing solutions examined for this dissertation, ranging from sixth to nineteenth, relied on traditional gradient-boosting techniques, primarily CatBoost and XGBoost, applied to manually engineered spectral and vegetation indices (ITU-GeoAI-Challenge, 2023). Despite evidence from the broader literature that such architectures can offer real, if specific, advantages, particularly in separating spectrally similar land-cover classes and in handling irregular, cloud-affected observation sequences, none of the public solutions seem to have attempted a deep learning architecture capable of learning temporal-spectral representations directly from the satellite time series. This raises a real, testable question instead of an assumption given a meagre budget of only 1,000 samples, does deep learning offer a quantifiable advantage over the established gradient-boosting baseline for this exact dataset and these exact regions, or does the additional architectural complexity fail to justify itself once the practical data constraints are taken into account, as at least one directly comparable arid-region study has suggested? The main issue this dissertation tackles is providing a response to this question rather than merely assuming an answer in either case.

## *1.2 Research aim and objectives*

### **Aim:**

In order to determine whether deep learning offers quantifiable advantages over an established gradient boosting baseline when the available training data is severely constrained, this dissertation will develop, implement, and critically evaluate machine learning and deep learning methods for annual cropland extent mapping at 10-metre resolution using multi-sensor satellite imagery over Iran as an in-depth, data-scarce, semi-arid case study region. Additionally, it will evaluate how the pipeline and findings could be scaled to the other two regions of the GeoAI challenge, Sudan and Afghanistan.

### **Objectives:**

1. To critically review the existing literature on machine learning and deep learning approaches to cropland and crop-type mapping from satellite imagery, with particular attention to multi-sensor data fusion, learning from limited labelled samples, and generalization across time periods.
2. To acquire and process multi-sensor satellite imagery (Sentinel-1, Sentinel-2 and Landsat-8) for Iran, and to engineer spectral and vegetation-index features consistent with the training-sample constraints of the original GeoAI Challenge.
3. To design, implement, and tune a classical gradient-boosting and ensemble baseline (CatBoost, Random Forest, and XGBoost) in line with the approaches used by the strongest publicly documented solutions to the original challenge.
4. To design and implement a deep learning model, specifically a temporal convolutional architecture, capable of learning directly from multi-temporal satellite sequences with minimal manual feature engineering.
5. To comparatively evaluate both approaches against a common evaluation protocol, using overall classification accuracy, per-class performance, and a targeted analysis of performance on spectrally similar cropland classes.

6. To evaluate the temporal extendibility of both approaches by testing performance on a time period distinct from the one used for training in Iran and to critically reflect on the practical and computational trade-offs between the two approaches.
7. To critically discuss the scalability of the pipeline developed for Iran to the GeoAI Challenge's other two regions, Sudan and Afghanistan, identifying what would transfer directly and what would require region-specific adaptation, without extending full implementation and evaluation to those regions.

### *1.3 Research approach*

This dissertation adheres to CRISP-DM (Cross-Industry Standard Process for Data Mining), which was chosen over the other data science approaches taken into consideration (KDD, SEMMA, and GABDO). This is because, in contrast to SEMMA, it treats business/research understanding as a first-class phase in its own right rather than folding it into data sampling and because of its explicit separation of business understanding, data understanding, data preparation, modelling, evaluation, and deployment. This technique is fully described in Chapter 3, along with how each step was interpreted for a research-only project without a live deployment target or human participants. The actual work completed under each phase is then presented in turn in Chapter 4. Since the dataset is made up only of remotely sensed satellite images and previously obtained, anonymised geographic point labels, no human subjects were used in this study at any point; the full ethical-approval stance resulting from this is outlined in Appendix A.

### *1.4 Dissertation outline*

The rest of this dissertation is structured as follows. After reviewing the pertinent literature, Chapter 2 identifies the gap this dissertation fills by going from the general field of deep learning in remote sensing to techniques that explicitly handle data scarcity and cross-temporal farmland mapping. The study approach is described in Chapter 3, along with the CRISP-DM technique used, its interpretation for this project, data access, and ethical considerations. The data analysis, including data comprehension, data preparation (collection and feature engineering), model creation for both the conventional and deep learning approaches, and their assessment, is presented in Chapter 4. Chapter 5 critically assesses the work done, addresses the scalability of the pipeline to Sudan and Afghanistan (Objective 7), and considers these findings in relation to the body of current literature and the goals of this dissertation. The dissertation is concluded in Chapter 6, which also includes personal thoughts on the project, a discussion of its limitations and future work, and a summary of its scholarly contributions. Appendix B includes supplementary information not needed by the main chapters, such as a more thorough requirement analysis of the project's design choices, whereas Appendix A deals with ethical approval.

## CHAPTER 2: LITERATURE REVIEW

### *2.1 Introduction*

The literature pertinent to the goals and objectives outlined in Chapter 1 is reviewed in this chapter. It is arranged thematically rather than chronologically, starting with broad foundational work on deep learning in remote sensing and progressing through classical machine learning approaches to cropland classification, deep learning architectures created especially for mapping crops and land cover, multi-sensor data fusion, and techniques for learning from limited labelled data and generalising over time. The chapter concludes with a summary of the data supporting deep learning in comparison to classical machine learning and a description of the gap in the literature that this dissertation fills.

### *2.2 Existing Global Cropland Products and Their Limitations*

There are currently several worldwide cropland and land-cover products available, such as WorldCover, GlobalLand30, and GFSAD30, usually at a resolution of 30 metres. Although these products have proven useful for coarse-scale monitoring, as described in Chapter 1, they share three limitations that inspired the GeoAI Challenge that forms the basis of this dissertation: inconsistent cropland definitions across products, measurable disagreement between products when compared directly over the same region, and infrequent updates that prevent year-to-year change detection (FAO, 2023). These constraints serve as the practical foundation for the literature reviewed in the rest of this chapter, all of which strive to produce cropland information that is higher resolution, more regularly updated, and more consistently defined than these current global products can provide.

### *2.3 Deep Learning in Remote Sensing: The Broader Landscape*

The most frequently cited thorough analysis of deep learning's use in remote sensing tasks in general is found in Zhu et al. (2017), who catalogue the application of convolutional neural networks, autoencoders, recurrent neural networks, and generative adversarial networks in land-cover classification, object detection, and scene understanding. Although their review predates much of the crop-specific work that follows, its main claim—that deep learning's ability to learn hierarchical spatial and spectral features directly from imagery rather than depending on manually created indices—remains the fundamental rationale for this dissertation's deep learning strand.

### *2.4 Classical Machine Learning Approaches to Cropland Classification*

In the initial 2023 GeoAI Challenge for Cropland Mapping, the best publicly available solutions relied solely on traditional gradient-boosting techniques. For example, the sixth-place public solution trained and compared XGBoost, AdaBoost, and CatBoost classifiers per country after engineering 181 spectral and vegetation indices per month from Sentinel-1, Sentinel-2, and Landsat-8 imagery across the three target countries. CatBoost consistently performed the best, achieving 0.9288 accuracy on the competition's public leaderboard (ITU-GeoAI-Challenge, 2023). This challenge is not the only one where CatBoost outperforms XGBoost on tabular, feature-engineered data. When comparing CatBoost, XGBoost, and LightGBM directly against a broader range of baselines (CART, AdaBoost, standard Gradient Boosting, Random Forest, Neural Networks, and SVM) for

global crop yield prediction, Mahesh and Soundrapandiyan (2024) found that CatBoost performed the best ( $R^2 = 0.9996$ , MAPE = 1.51), followed closely by XGBoost ( $R^2 = 0.995$ ), and LightGBM was noticeably less consistent across dataset configurations ( $R^2$  as low as 0.33 in some settings). Although the underlying task, tabular yield regression rather than imagery-derived classification, is different from this dissertation's, this difference between CatBoost and its closest gradient-boosting competitors, verified by formal statistical significance testing (Diebold-Mariano test), provides independent, peer-reviewed support to the pattern seen in the challenge's own public solutions.

Beyond this particular difficulty, traditional machine learning has a solid track record in the classification of crops in general. In the Haouz plain, a semiarid region of Morocco with an average annual rainfall of only 250 mm, Moumni and Lahrouni (2021) used a fusion-based classical machine learning approach (Maximum Likelihood, SVM, and a shallow Artificial Neural Network) for crop-type mapping. This climate context is very similar to the Afghanistan, Iran, and Sudan regions examined in this dissertation. Without utilising deep learning, combining Sentinel-2 optical and Sentinel-1 radar data increased overall accuracy from 85.83% (optical alone) to 89% and Kappa from 0.81 to 0.85, thereby supporting the multi-sensor fusion approach used in this research (Section 2.6). The aggregate body of work demonstrates the validity of the gradient-boosting baseline used in Chapter 4 of this dissertation: it is a robust, well-supported strategy in and of itself rather than a weak strawman method selected for the deep learning comparison to be easily defeated.

### *2.5 Deep Learning Architectures for Crop and Cropland Mapping*

Deep learning architectures designed for satellite image time series classification have been developed by a different body of research, which is mainly unrelated to the previously mentioned traditional gradient-boosting paradigm. Instead of proposing a single fixed design, Pelletier, Webb, and Petitjean (2019) conducted the first comprehensive, systematic study of the architecture (79 architectures across 2,000 trained models) and introduced the Temporal Convolutional Neural Network (TempCNN), which applies one-dimensional convolutions along the temporal axis of a satellite time series rather than across spatial dimensions. Their base TempCNN (three convolutional layers of 64 units, filter size 5, dropout 0.5, batch normalisation) achieved 93.45% overall accuracy against Random Forest's 90.92% and an RNN's 91.87% on a dataset of approximately one million labelled Formosat-2 pixel time series from southwest France. This margin was statistically significant but modest (paired t-test,  $p < 0.001$  against RF) and was not evenly distributed across all thirteen classes examined. This dissertation adopts this finding as its main testable hypothesis, which is that deep learning's advantage specifically lies in discriminating confusable classes rather than in a uniform accuracy improvement. Notably, the same study found that pooling layers, a nearly universal component in computer-vision CNNs, actively harmed accuracy for this task, attributed to the temporal location and amplitude of phenological signals being informationally critical rather than safely discardable.

The same issue has been tackled from several perspectives by alternative architectures. In order to achieve multi-temporal Sentinel-2 land-cover classification across 17 crop classes in the vicinity of Munich, Germany (89.5-89.7% overall accuracy, Kappa 0.870), Rußwurm and Körner(2018) used bidirectional convolutional LSTM/GRU recurrent encoders. They purposefully used unfiltered, cloud-contaminated imagery instead of pre-masking clouds. By visualising internal LSTM cell activations, they were able to pinpoint particular cells whose gate activity abruptly decreased during cloudy observations. This demonstrated that the network had learned an implicit cloud-filtering behaviour solely

from data, without receiving explicit cloud labels, which is a genuinely useful simplification over traditional cloud-masking pipelines. Ji et al. (2018) extended convolutional approaches into three dimensions, modelling spatial and temporal information jointly rather than as separate one-dimensional signals, and provide the clearest quantified precedent in the reviewed literature for the "spectrally similar classes" hypothesis: on a 2016 Chinese Gaofen-2 dataset where a local upland rice variety and corn were near-spectrally-identical for most of the growing season, their 3D CNN reduced the rice-corn confusion-matrix entry from 0.12 (2D CNN) to 0.02, lifting Kappa from 0.820 to 0.924, while showing only a marginal advantage (0.4-0.6 accuracy points) on a separate, spatially-separable 2015 dataset where classes were not confusable in the first place. Additionally, the same study showed that an active-learning sample-selection strategy reduced the labelled data required to achieve 90% accuracy from 1,500 (random sampling) to 400 points, a roughly 3.75-fold efficiency gain that is directly applicable to any pipeline operating under a hard cap on labelled training samples, including the pipeline used in this dissertation. More recently, Metzger et al. (2022) addressed a particular practical challenge—irregular, cloud-gapped observation sequences—that any application utilising Sentinel-2 imagery over Afghanistan, Iran, or Sudan will also encounter. Benchmarked against LSTM, GRU, TempCNN, and Transformer baselines, their Neural Ordinary Differential Equation (ODE-RNN) hybrid demonstrated its greatest accuracy gains precisely in the scenario with the fewest available observations, i.e., the most cloud-affected sequences, providing a plausible, if architecturally more complex, path to greater robustness in precisely the conditions this dissertation's target region presents. The Lightweight Temporal Attention Encoder (L-TAE), a computationally efficient self-attention mechanism created especially for large-scale satellite time series processing, was first presented by Sainte Fare Garnot and Landrieu (2020). In Sainte Fare Garnot and Landrieu (2021), it was combined with multi-scale spatial convolution to create U-TAE, which was assessed using their own PASTIS benchmark (2,433 Sentinel-2 sequences across four French regions, 18 crop classes). While using comparable or fewer trainable parameters, U-TAE outperformed six reimplemented competing SITS encoders, including 3D-Unet and several ConvLSTM variants, on both accuracy (83.2% vs. 3D-Unet's 81.3%) and mIoU (63.1 vs. 58.4). Ablating the temporal dimension completely (single-date input) cost 24.8-42.5 accuracy points, a striking, quantified demonstration of how much the temporal dimension itself contributes regardless of architecture. It was discovered that the same model automatically down-weighted cloud-affected observations by 58% in its attention scores. This is a second, independently obtained (attention-based rather than LSTM-gate-based) confirmation of the implicit cloud-robustness Rußwurm and Körner (2018) observed, indicating that this is a true, architecture-agnostic property of temporal deep learning models rather than an artefact of one particular design.

In this dissertation, these architectures collectively constitute the deep learning option compared to the classical baseline. The temporal convolutional approach of Pelletier, Webb, and Petitjean (2019) is chosen as the main deep learning model for implementation because it is the most thoroughly validated of the architectures examined and because it is a more practical option within the time and computational constraints of an MSc dissertation due to its lower architectural complexity compared to attention-based, recurrent, or Neural ODE approaches. However, Pelletier, Webb, and Petitjean's (2019) reported TempCNN-over-RF margin is treated in this dissertation as an upper bound on what might be expected at Iran's much smaller training scale, not as a result to be replicated, because their own dataset (roughly one million labelled samples) is roughly three orders of magnitude larger than the training budget available under the GeoAI Challenge's constraint (1,000 samples per region).

## 2.6 Multi-Sensor Data Fusion

The integration of several satellite sensors instead than depending on a single data source is a recurring theme in the deep learning and classical literature discussed above. Sentinel-1 radar imagery, Sentinel-2 optical imagery, and Landsat-8 data were combined in all public solutions to the 2023 GeoAI Challenge instead of using any one sensor alone. This is indicative of a wider understanding that radar and optical sensors capture complementary information: radar is unaffected by cloud cover but less directly interpretable in terms of vegetation biochemistry, while optical imagery provides intuitive vegetation indices like NDVI but is useless during cloud cover (ITU-GeoAI-Challenge, 2023). While the wider multi-sensor fusion pattern seen in the competition solutions is consistent with the general remote sensing literature reviewed by Zhu et al. (2017), Singh et al. (2022) show the usefulness of Sentinel-2 specifically for deep-learning-based agricultural land-use mapping. For both the classical and deep learning models used in Chapter 4, this dissertation uses the same multi-sensor fusion technique because deviating from this proven method will add an unwarranted and detrimental confound to the comparison.

## 2.7 Learning from Limited Labelled Data

The 1,000 labelled training sample limit per region set by the GeoAI Challenge is not a cap. It reflects a real, well-documented issue in the larger literature: collecting geo-referenced ground-truth labels for cropland is costly and time-consuming. This is particularly true in the semi-arid, historically under-surveyed, and sometimes conflict-affected areas where precise cropland maps would be most beneficial. With the EuroCropsML benchmark, which was created especially to assess few-shot crop-type classification techniques under precisely this type of constrained-label setting, Reuss et al. (2025) directly address this issue. Their work offers a helpful methodological benchmark for assessing models under the equivalent constraint imposed by the GeoAI Challenge. The most directly pertinent study found in this review, however, is Gajardo et al. (2025), who examined cropland mapping in Nigeria, a nation that faces data scarcity issues that are similar to those in Sudan and Afghanistan in this dissertation. The study was framed as binary crop-versus-non-crop classification. They compared a Random Forest classifier against single- and multi-headed Long Short-Term Memory networks using Google Earth Engine to extract Sentinel-1, Sentinel-2, ERA5 climate, and SRTM elevation data for 1,827 manually labelled points. They discovered that adding local (Nigeria) training data on top of global crowdsourced (Geowiki) data improved Random Forest accuracy from 0.591 to 0.837 (a gain of 0.664 to 0.842 (a gain of 0.178). An early indication that deep learning architectures may provide particular benefits in precisely the low local-data conditions this dissertation examines is their multi-headed LSTM, which routes local and global samples through distinct classification heads sharing one LSTM backbone. This architecture was the only one to benefit from additional global data, particularly when local data was scarce. Notably, however, an existing worldwide solution called ESA WorldCover 2020, which was based on a CatBoost model ( $F1 = 0.825$ ) trained on a far bigger, although less regionally relevant, sample, outperformed their own best-performing model (the single-headed LSTM,  $F1 = 0.814$ ) on the same Nigeria test set. This is a recent (2025), independently obtained example of the same result this dissertation is intended to test for Iran: a specially designed deep learning model that was trained on a small amount of local data lost to an established traditional gradient-boosting baseline.

## *2.8 Temporal Generalisation and Cross-Year Transferability*

Beyond precise mapping in a single period, the original GeoAI Challenge's second stated goal was to evaluate the temporal extendibility of suggested techniques, i.e., whether a model trained on data from one time period continues to generalise when applied to a different period in the same region (Zindi, 2023). Since land use, cropping patterns, and even the availability of particular satellite imagery vary from year to year, this aspect of the reviewed literature is relatively understudied in comparison to single-period classification accuracy, but it directly relates to the practical value of any cropland-mapping method. Cross-scale sample transfer for high-resolution cropland extraction is a closely related problem that Zhang et al. (2023) address. They offer a knowledge-driven automated solution for transferring labelled samples across different spatial scales rather than time periods specifically, but this dissertation conceptually addresses the same problem—that a model's learned representation must generalise beyond the precise conditions under which it was trained. In order to directly address this relatively understudied dimension, the sixth objective of this dissertation evaluates both the classical and deep learning models against a time period different from their training data.

## *2.9 Deep Learning versus Classical Machine Learning: Synthesis of the Comparative Evidence*

After examining each tradition separately, it is worthwhile to compile the data that most directly addresses the main question of this dissertation: does deep learning provide a quantifiable advantage over classical gradient boosting for cropland classification under a limited training-sample budget, in a semi-arid, data-scarce context?

This dissertation views the really conflicting findings as a strength rather than a drawback of the study issue. Using Sentinel-2 time series, Mohammadi Mobarakeh, Pourmanafi, and Ahmadi (2025) precisely classified complex multi-crop systems in central Iran, one of the dissertation's own target regions. This is likely the most directly relevant piece of comparison information available. They reported Kappa coefficients of 0.90 for Random Forest, 0.86 for XGBoost, 0.82 for SVM, and 0.89 for the Temporal CNN before noise-point removal from the training data (rising to 0.95, 0.94, 0.88, and 0.94 respectively after their noise-point-removal quality-control step). They compared Random Forest, XGBoost, Support Vector Machines, and a Temporal CNN. Although Random Forest was by far the best overall, the Temporal CNN's unique advantage was in more consistently separating spectrally similar crop types that the classical methods occasionally confused, despite its significantly higher computational cost and complexity. This was in line with Pelletier, Webb, and Petitjean's (2019) original finding and independently confirmed by Ji et al.'s (2018) rice-corn result (Section 2.5). The Nigerian study by Gajardo et al. (2025) provides a complementary and more starkly quantified finding: adding local training data increased Random Forest accuracy by 0.246 and LSTM accuracy by 0.178; the multi-headed LSTM variant was the only one that could benefit from global training data in its absence; however, their best deep learning model was ultimately outperformed on their own test set by an existing classical (CatBoost-based) product, WorldCover.

When considered collectively, this comparative evidence indicates that a straightforward binary response to the study issue of this dissertation is implausible. Under the training-sample constraints this dissertation imposes, deep learning does not seem to guarantee an overall accuracy improvement over well-tuned gradient boosting. This is consistent with the actual result of the original 2023 challenge, whose top public solutions were all classical. The literature does, however, regularly highlight a more focused and limited benefit of deep learning: improved resilience in the absence or scarcity of local training

data, as well as improved discriminating of spectrally identical or otherwise confusing classes. This is the precise, falsifiable hypothesis this dissertation is designed to test in Chapters 4 and 5, rather than broader and less answerable question of whether deep learning is simply better than classical machine learning in general.

### *2.10 Gap Analysis and Positioning of the Current Study*

Drawing the threads of this chapter together, three specific gaps in the existing literature can be identified.

First, the vast bulk of the literature on crop mapping—both classical and deep learning—is created and verified in data-rich, well examined environments, primarily in Western Europe and North America. There are studies that concentrate on semi-arid, data-scarce contexts similar to Afghanistan, Iran, and Sudan, like Moumni and Lahrouni (2021) in Morocco and Gajardo et al. (2025) in Nigeria, but they are still rare in the field overall and do not directly relate to the regions or dataset of this dissertation.

Second, and more specifically, no publicly documented solution to the original 2023 GeoAI challenge for cropland mapping appears to have implemented a deep learning architecture at all. Every top-placing public solution identified for this dissertation relied on classical gradient boosting. This represents a direct, addressable gap: a deep learning baseline simply doesn't yet exist for this specific dataset, these specific regions and this specific training sample constraint, even though the wider literature reviewed in Section 2.5 gives good reason to expect that deep learning might offer a genuine advantage.

Third, despite its obvious practical significance for any cropland-mapping method meant for real-world use rather than a one-time academic exercise, temporal extendibility—the second objective of the original challenge—is relatively under-explored in comparison to single-period classification accuracy across the literature reviewed in this chapter.

To directly address all three gaps, this dissertation explicitly assesses the temporal extendibility of both models rather than relying solely on single-period accuracy. It does this by implementing and comparing a deep learning model against a classical gradient-boosting baseline on the precise dataset, regions, and training-sample constraint specified by the 2023 GeoAI Challenge.

### *2.11 Summary*

The literature pertinent to cropland extent mapping from satellite imagery has been reviewed in this chapter. From the general field of deep learning in remote sensing, through traditional gradient-boosting techniques and deep learning architectures created especially for crop classification, to the more focused issues of multi-sensor fusion, limited labelled data, and temporal generalisation. The chapter has identified a specific, addressable gap: no deep learning baseline exists for the 2023 GeoAI Challenge for Cropland Mapping dataset, and the comparative evidence from closely related contexts suggests deep learning's likely advantage lies in discriminating spectrally similar classes and in resilience under data scarcity rather than in a uniform accuracy improvement. Chapter 3 now turns to the research approach adopted to investigate this question.

## CHAPTER 3: RESEARCH APPROACH

### 3.1 Introduction

The research approach used for this dissertation is described in this chapter. The selection of CRISP-DM is supported by a review of the data science approaches taken into consideration in Section 3.2. The six phases of CRISP-DM are mapped onto this particular project in Section 3.3. The tooling and development environment used throughout, as well as the data access path, are described in Section 3.4. Ethical considerations are covered in Section 3.5. Chapter 4 then carries out the work this chapter sets out a plan for.

### 3.2 Choice of Methodology

The following four data science research approaches were taken into consideration: GABDO, SEMMA (Sample, Explore, Modify, Model, Assess), CRISP-DM (Cross-Industry Standard Process for Data Mining), and KDD (Knowledge Discovery in Databases). Although their emphasis is different, they both organise a data science project as a series of steps from raw data to an assessed, deployable model. SEMMA begins with sampling and is comparatively silent on the upstream question of what the project is actually for; KDD emphasises the interpretation and knowledge-extraction step at the end of the process more than the business or research framing at the start. CRISP-DM was chosen for this dissertation because its six phases—Business Understanding, Data Understanding, Data Preparation, Modelling, Evaluation, and Deployment—map neatly onto the actual order of work completed for this dissertation and because it treats Business Understanding (in this case, research understanding: the aim, objectives, and the specific gap in the literature this project addresses) as a distinct, first-class phase rather than an implicit precondition.

### 3.3 CRISP-DM Applied to This Project

Business Understanding: Translated for a research dissertation rather than a commercial project, this phase aligns with the research problem, aim, and objectives outlined in Chapter 1: evaluating how any resulting pipeline could scale to Sudan and Afghanistan and determining whether deep learning offers a quantifiable advantage over classical gradient boosting for cropland classification under the training-sample constraint of the GeoAI Challenge. In order to understand the current state of the art for this particular dataset before any new work started, this phase also involved reviewing the public solutions to the original 2023 challenge and the existing literature (Chapter 2) to confirm the research question was truly open rather than already answered.

Data Understanding: Prior to any feature engineering, this phase entailed examining the GeoAI Challenge's own dataset documentation and guidelines, verifying that the dataset was still directly downloadable, and examining the downloaded files to determine their true structure and content (point coordinates and binary labels only, no satellite-derived features). The competition's own beginning notebook was also critically examined, revealing a real coordinate-order flaw that is fully detailed in Chapter 4.

Data Preparation: In this phase, the shared multi-region dataset was filtered down to Iran's working subset, and the entire multi-sensor feature-extraction pipeline was carried

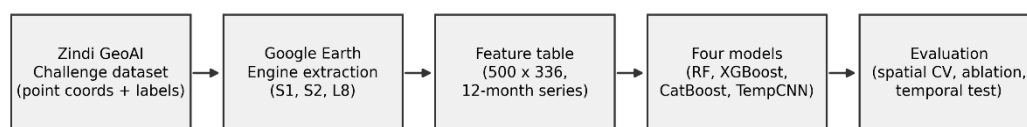
## CLASSICAL MACHINE LEARNING VERSUS DEEP LEARNING FOR DATA-SCARCE CROPLAND MAPPING: A CASE STUDY OF IRAN

out. This included using Google Earth Engine to obtain Sentinel-1, Sentinel-2, and Landsat-8 imagery, engineering spectral and vegetation indices, creating a twelve-month time series per point, and applying data quality checks to the final feature table prior to modelling.

**Modelling:** This phase covered the design, implementation, and hyperparameter tuning of both the classical baselines (Random Forest, XGBoost, and CatBoost) and the deep learning model (TempCNN), including, as described in full in Chapter 4, an iteration in which a genuine error in the deep learning model's evaluation methodology was identified and corrected.

**Evaluation:** This phase covered the comparative evaluation methodology, spatially blocked cross-validation, statistical significance testing, the WorldCover-based substitute analysis for spectrally similar surfaces, and the temporal extendibility test, together with the results these produced.

**Deployment:** This dissertation is an academic comparative study rather than an operational system, so no live deployment was carried out; this phase is instead addressed as the scalability discussion in Chapter 5 (Objective 7), considering what would be required to extend the evaluated pipeline to Sudan and Afghanistan without carrying out that extension in full.



### 3.4 Data Access and Development Environment

**Data access:** The dataset came from the 2023 ITU/FAO/UNODC GeoAI Challenge for Cropland Mapping, which was held on Zindi (Zindi, 2023). As of this writing, it is still possible to download Train.csv, Test.csv, SampleSubmission.csv, and a starter notebook straight from the competition page without having to rebuild the dataset from Earth Engine from scratch. This was a data-access issue that was settled early in the project and verified in practice prior to implementation. Reuse for research and teaching is specifically permitted by the dataset's CC-BY-SA 4.0 license.

**Development environment:** Python 3.11 was used for the implementation. Google Earth Engine (Gorelick et al., 2017), a cloud-based platform for planetary-scale geospatial analysis that eliminates the need to download and locally store raw satellite scenes, was used to access and process satellite imagery. Feature extraction queries were executed under Earth Engine's non-commercial Community usage tier, which was adequate for the point-based, single-year extraction workload this project required. CatBoost (Prokhorenkova et al., 2018), XGBoost (Chen and Guestrin, 2016), and the RandomForestClassifier from scikit-learn (Pedregosa et al., 2011) were used to create classical models; PyTorch was used to implement the deep learning model (Paszke et al., 2019). Pandas and NumPy were used throughout the data handling process. In accordance with the repeatability requirements of this dissertation (Section 3.5), all tools utilised are free and open source.

### *3.5 Ethical Considerations*

No human subjects were used in this study; the dataset is made up only of remotely sensed satellite imagery and previously gathered, anonymised geographic point coordinates and binary labels that do not contain any personally identifying information. There was no survey, interview, or interaction with participants, nor was any new data gathered from human subjects. The GeoAI Challenge dataset is specifically licensed for research use, while the satellite datasets utilised (Sentinel-1, Sentinel-2, Landsat-8, and the ESA WorldCover product used in Chapter 4) are all freely and publicly accessible under open licenses. Because of this, it was determined that this experiment did not need formal ethical approval; Appendix A has the complete statement that supports this conclusion.

A separate, methodological form of research integrity is addressed throughout this dissertation rather than confined to this section: reproducibility. In keeping with the transparency this dissertation maintains throughout, every design decision—including the two technical flaws discovered and fixed during data preparation (Chapter 4) and the more significant evaluation error discovered and fixed during modelling (also Chapter 4)—is sufficiently detailed to allow for an audit or independent replication of the work.

### *3.6 Summary*

This chapter outlines CRISP-DM as the methodology used for this dissertation, justifies it against the alternatives taken into consideration, and maps its six phases onto the data, models, assessment strategy, and real research challenge of this particular project. It has also discussed the development environment utilised throughout, the data access route verified for this project, and the ethical issues raised by a study that was deemed to not require formal ethical approval because it did not involve any human subjects. The processes of Data Understanding, Data Preparation, Modelling, and Evaluation are now fully carried out in Chapter 4.

## CHAPTER 4: DATA ANALYSIS

### 4.1 Introduction

The work completed under CRISP-DM's Data Understanding, Data Preparation, Modelling, and Evaluation phases is presented in this chapter using the methods described in Chapter 3. Data Understanding: The GeoAI Challenge dataset and its structure is covered in Section 4.2. Data preparation, including multi-sensor feature engineering and applied data quality checks, is covered in Section 4.3. Modelling is covered in Section 4.4, which includes the deep learning model, the classical baselines, and a clear explanation of a real methodological error that was discovered and fixed in between two modelling iterations. Evaluation and the comparative outcomes it created are covered in Section 4.5. The results are then critically examined in Chapter 5 in relation to the goals of this research and the body of current literature.

### 4.2 Data Understanding

#### 4.2.1 The GeoAI Challenge dataset

Each of the 1,500 rows in `Train.csv` and `Test.csv` include a point identifier, latitude, longitude, and, for the training set only, a binary cropland/non-cropland label. These are ground-truth point locations that cover all three of the challenge's areas (Afghanistan, Iran, and Sudan) without a specific region column; no satellite-derived features are included.

#### 4.2.2 Regional filtering

Latitude/longitude thresholds based on the approximate geographic area of each country were used to assign points to regions: Afghanistan where latitude  $\geq 25^\circ$  and longitude  $\geq 60^\circ$ , Iran where latitude  $\geq 25^\circ$  and longitude  $< 60^\circ$ , and Sudan where latitude  $< 25^\circ$ . In accordance with this dissertation's choice to focus exclusively on Iran (Section 1.1), applying this rule confirmed an exactly even split of 500 points per region in both files. Each region's training points perfectly balanced 250 cropland / 250 non-cropland, isolating a working Iran subset of 500 labelled training points and 500 unlabelled test points.

#### 4.2.3 A data-quality issue in the provided starter material

Instead of being accepted without question, the competition's beginning notebook was examined before any of its code was reused. Because it affects the validity of any baseline created directly on top of the unaltered starter code, this review found a functional flaw that is worth documenting. The feature-extraction function in the notebook creates an Earth Engine point geometry as `ee.Geometry.Point(lon, lat)`, which anticipates longitude first. Nevertheless, every extracted point is geolocated incorrectly because the two local variables feeding this call are assigned from the source columns in reverse order (`lon = row['Lat']`, `lat = row['Lon']`). Additionally, the imagery pipeline in the starter notebook was restricted to four raw Sentinel-2 bands without vegetation indices, no radar or Landsat-8 data, a single whole-period median composite instead of a temporal sequence, and its date range was hardcoded to Afghanistan's reference period (April-May 2022) instead of Iran's (July 2019-June 2020). In the workflow outlined below, no code from the beginning notebook was used

again without modification.

```
def extract_mean_pixel_values(row):  
    lon = row['Lat']  
    lat = row['Lon']  
  
    # Create a point geometry for the labeled location  
    point = ee.Geometry.Point(lon, lat)
```

#### 4.3 Data Preparation

##### 4.3.1 Sensors and reference period

Three satellite sensors were used: Sentinel-2 (optical), Sentinel-1 (synthetic aperture radar), and Landsat-8 (optical and thermal), in accordance with the multi-sensor fusion pattern established across both the conventional and deep learning literature covered in Section 2.6. In order to provide the deep learning model with an actual time series to learn from, features were extracted for Iran's reference period, July 2019 to June 2020, and divided into twelve monthly composites rather than a single whole-period composite (Section 4.4); the classical models use the same monthly features flattened into a single wide row.

##### 4.3.2 Bands, indices, and cloud masking

Ten spectral bands (B2, B3, B4, B5, B6, B7, B8, B8A, B11, B12) and five derived vegetation and water indices (NDVI, EVI, SAVI, NDWI, and NDMI) were extracted each month for Sentinel-2. Before compositing, pixels impacted by clouds were eliminated using the QA60 bitmask. VV and VH backscatter were retrieved for Sentinel-1 along with a cross-polarization ratio (VH – VV). Six optical bands (SR\_B2-SR\_B7) and one thermal band (ST\_B10) were extracted from Landsat-8, cloud-masked using the QA\_PIXEL bitmask, and three additional indices (NDVI, NDWI, and NDMI) were calculated using the official Collection 2 Level 2 scale factors. Each point contains 336 feature values in total, which are reproduced over a 12-month period and include 15 Sentinel-2 features, 3 Sentinel-1 features, and 10 Landsat-8 features.

Instead of issuing one server round-trip per point as the starter notebook's method would have required, extraction was implemented using Earth Engine's `reduceRegions` operation, applied once per month per sensor across the entire set of 500 points simultaneously. This reduced the extraction workload from several hundred separate Earth Engine API calls to 72 (12 months × 3 sensors × 2 data splits).

An initial iteration of this extraction procedure narrowed the collection to the pertinent month after applying Earth Engine's cloud-masking and index-computation capabilities to each sensor's complete, globally unfiltered image collection. Because Earth Engine creates a computation graph sized to the entire, all-time global archive when functions are mapped before the collection is filtered, even though only a few sample points are ultimately evaluated, chaining two mapped functions together in this order caused the same query to hang indefinitely, even though a single mapped function applied in this manner executed quickly. This was fixed by rearranging the pipeline to filter by date first and then apply the mapping functions to the resulting tiny monthly subset.

```
s2_img = (  
    ee.ImageCollection("COPERNICUS/S2_SR_HARMONIZED")  
    .filterDate(start, end)  
    .map(mask_s2_clouds)  
    .map(add_s2_indices)  
    .mean()  
)
```

#### 4.3.3 Missing data

Not every point has a valid observation for every sensor in every month. Two months (December 2019 and April 2020) had significantly higher missing-value rates due to Landsat-8's single-satellite, sixteen-day revisit cycle and seasonal cloud cover. In the second extracted year (July 2020-June 2021, used for the temporal extendibility test in Section 4.5.5), one month (November 2020) had no usable Landsat-8 acquisitions for any point in the study area. Sentinel-1 and Sentinel-2 were relatively unaffected. This is fixed at the modelling stage: CatBoost, XGBoost, and RandomForest were given features with missing values handled by native support or median imputation as appropriate, and the deep learning model imputes each missing channel-month with its training-set median before standardisation. This represents a real data-availability limitation of the underlying satellite archive rather than a flaw in the extraction pipeline..

#### 4.3.4 Data quality checks

Prior to modelling, four checks were performed against the extracted feature table. Five out of 162,630 non-missing values (0.003%) fell outside of physically plausible ranges for their respective quantity; these were isolated edge cases in vegetation index or thermal-band columns and were left unaltered due to their insignificant prevalence. Out of the 500 points, no exact duplicate feature rows were discovered. As would be expected given that they are derived from a common set of underlying spectral bands, same-month engineered vegetation indices were found to be highly correlated with one another (for instance, EVI and SAVI in the same month correlate at  $r \approx 0.998$ ); this has no direct bearing on model training but is pertinent to the interpretation of feature-importance results (Section 4.5.2). Lastly, it was discovered that both the classical and deep learning models independently misclassified eleven of the 500 points (2.2%) out of fold; this is reported as a candidate list for manual review against the original point locations and imagery, not as a claim that these particular labels are incorrect.

### 4.4 Modelling

#### 4.4.1 Classical baselines

Three classical models were used: Random Forest and XGBoost were added to expand the classical comparison set toward the one used by the closest published benchmark study (Mohammadi Mobarakeh, Pourmanafi, and Ahmadi, 2025), which compares Random Forest, XGBoost, and SVM against a temporal CNN; CatBoost was consistent with the strongest publicly documented solutions to the original challenge (Section 2.4). A scope limitation covered in Chapter 5 was the absence of a Support Vector Machine baseline.

Using the spatially blocked cross-validation technique outlined in Section 4.5.1, all three models were cross-validated after being modified via randomised hyperparameter search (25 iterations per model) over model-appropriate parameter distributions. After an early untuned test revealed the model converged after two to three boosting iterations on a training set of this size, CatBoost's iteration range was purposefully kept modest (20–200) instead of a wider range initially tried (200–1,000).

#### 4.4.2 Deep learning model: architecture

The deep learning model is based on Pelletier, Webb, and Petitjean's (2019) Temporal Convolutional Neural Network architecture, which consists of three one-dimensional convolutional layers, batch normalisation, a ReLU activation, dropout, a fully connected layer, and a single output unit for binary classification. Since the original paper's dataset (roughly one million labelled samples) is roughly three orders of magnitude larger than the 500 samples available here, the base configuration (three convolutional layers of 64 filters, kernel size 5, dropout 0.5, dense layer of 256 units) was used as the center point of the tuning search instead of being fixed outright. The input to the network is a tensor of shape (28 channels, 12 timesteps) per point, the 28 channels corresponding to the 15 Sentinel-2, 3 Sentinel-1, and 10 Landsat-8 features described in Section 4.3.2, arranged as a genuine monthly sequence.

```
self.conv = nn.Sequential(  
    nn.Conv1d(n_channels, n_filters, kernel_size, padding=pad),  
    nn.BatchNorm1d(n_filters),  
    nn.ReLU(), nn.Dropout(dropout),  
    nn.Conv1d(n_filters, n_filters, kernel_size, padding=pad),  
    nn.BatchNorm1d(n_filters),  
    nn.ReLU(), nn.Dropout(dropout),  
    nn.Conv1d(n_filters, n_filters, kernel_size, padding=pad),  
    nn.BatchNorm1d(n_filters),  
    nn.ReLU(), nn.Dropout(dropout),  
)  
self.classifier = nn.Sequential(  
    nn.Flatten(), nn.Linear(n_filters * n_timesteps, dense_units), nn.ReLU(),  
    nn.Dropout(dropout), nn.Linear(dense_units, 1),  
)
```

A random search over ten configurations was conducted, varying the number of convolutional filters (32/64/128), kernel size (3/5/7), dense layer width (128/256), dropout rate (0.3/0.5/0.7), learning rate ( $1 \times 10^{-3}$ / $5 \times 10^{-4}$ / $1 \times 10^{-4}$ ), and weight decay ( $0/1 \times 10^{-4}/1 \times 10^{-3}$ ), training with the Adam optimiser and a batch size of 32, consistent with the original paper's setup.

#### 4.4.3 Deep learning model: Iteration 1

The initial modelling iteration determined which training epoch's weights to keep by comparing the model to the held-out cross-validation fold after each of its (up to 100) training epochs and keeping the epoch that scored the highest on that same fold. TempCNN outperformed all three conventional models with a tuned five-fold cross-validated Kappa of  $0.960 \pm 0.022$  under this technique (Section 4.5.1, Table 4.1).

This outcome was not taken at face value. Before it was written up, two checks were performed. Given that the 500 points are not spatially uniform, the first question is whether the cross-validation folds were spatially independent of one another. The median nearest-neighbor distance between points is roughly 490 meters, and grouping points within 500 meters of one another collapses the 500 points into 342 distinct spatial groups, 93 of which contain more than one point. This poses a serious risk because points this close together can share nearly identical satellite signatures (Roberts et al., 2017). Rerunning cross-validation using `GroupKFold`, grouped on these spatial clusters such that no group is ever split between training and test, yielded results that were within 0.01-0.02 Kappa of the initial random split, eliminating this as a possible explanation. Second, the raw out-of-fold error counts that resulted from doing a temporal extendibility test and a land-cover-based error analysis (Section 4.5, Section 4.5.4) using the Iteration 1 models for consistency did not mathematically match the accuracy that the 0.960 Kappa figure suggested.

By comparing the model to the same fold that it would ultimately be judged on at each epoch and retaining the best-scoring epoch, which cherry-picks the best of up to 100 attempts tested against the evaluation data itself, the true source of this mismatch was found. Although this is leakage into model *selection* rather than directly into the model's learnt weights, its impact on the reported score is comparable to more well-known types of test-set leakage.

#### 4.4.4 Deep learning model: Iteration 2

After training and epoch selection are finished, the evaluation function was changed so that best-epoch selection uses an inner validation split carved from each fold's training section only; the fold that is being scored is used exactly once. The hyperparameter search was re-run completely under the corrected procedure instead of just modifying the final reported score because the Iteration 1 evaluation function was also used inside the hyperparameter search, where it slightly double-counted the same validation split for both epoch selection and configuration selection.

```
Python
X_tr, X_val, y_tr, y_val = train_test_split(
    X_train, y_train, test_size=0.15, random_state=SEED, stratify=y_train
)
# [... standard training loop over epochs, using X_tr/y_tr only ...]

model.eval()
with torch.no_grad():
    val_pred = (torch.sigmoid(model(X_val_t)) >
0.5).float().numpy().astype(int)
    val_kappa = cohen_kappa_score(y_val, val_pred)
    if val_kappa > best_val_kappa:
        best_val_kappa = val_kappa
        best_state = {k: v.clone() for k, v in model.state_dict().items()}

# X_test is touched exactly once, after the best epoch is already fixed:
model.load_state_dict(best_state)
model.eval()
with torch.no_grad():
    test_pred = (torch.sigmoid(model(X_test_t)) > 0.5).float().numpy().astype(int)
```

Note that `X_val_t`, not `X_test_t`, is what the epoch-selection loop checks. The fold

actually being scored ( $X_{test\_t}$ ) only appears once, after the best epoch is already fixed.

Reversing the ranking from Iteration 1, TempCNN's tuned five-fold cross-validated Kappa under Iteration 2 was  $0.920 \pm 0.013$ , the lowest of the four models. In contrast to the configuration chosen in Iteration 1 (`kernel_size=7`, `weight_decay=0.0`), the revised optimal configuration was `n_filters=64`, `kernel_size=5`, `dense_units=256`, `dropout=0.5`, `lr=1e-3`, `weight_decay=1e-3`. Every TempCNN result provided in Section 4.5 and throughout Chapter 5 is generated using the process of Iteration 2, which is detailed above.

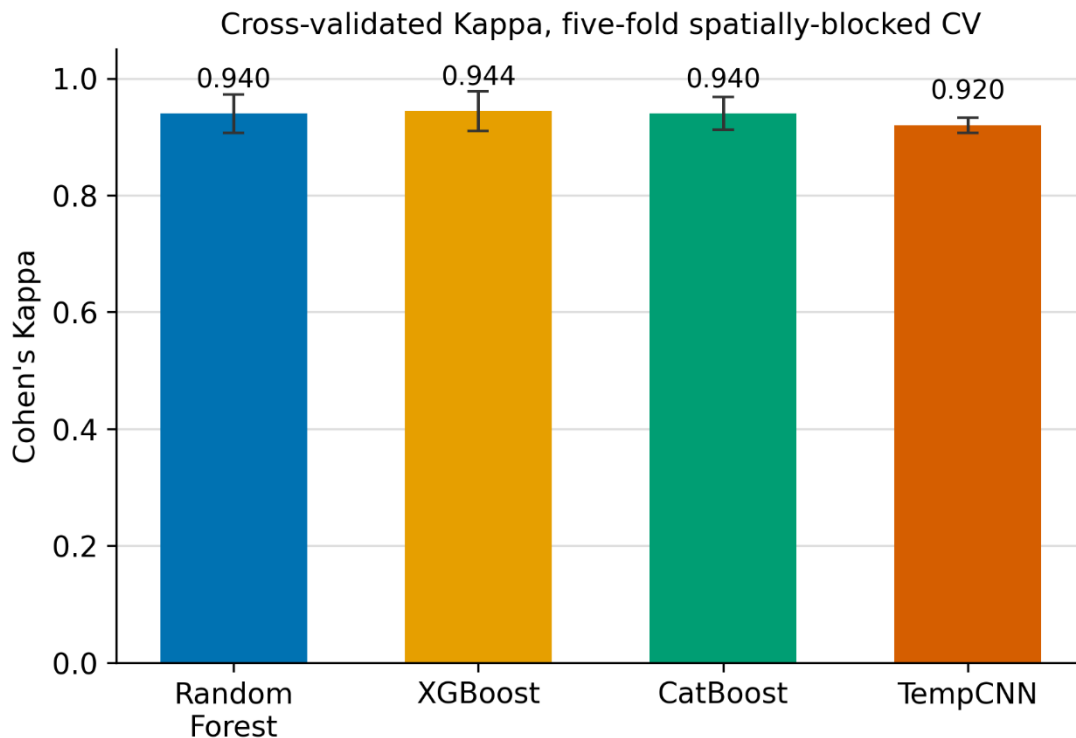
#### 4.5 Evaluation

##### 4.5.1 Comparative classification performance

Table 4.1 reports each model's cross-validated Cohen's Kappa, using spatially-blocked five-fold cross-validation (GroupKFold on the 342 spatial groups identified in Section 4.4.3), with tuned hyperparameters for all four models.

| Model                 | Kappa             |
|-----------------------|-------------------|
| Random Forest         | $0.940 \pm 0.033$ |
| XGBoost               | $0.944 \pm 0.034$ |
| CatBoost              | $0.940 \pm 0.028$ |
| TempCNN (Iteration 2) | $0.920 \pm 0.013$ |

**Table 4.1: Cross-validated Kappa, five-fold spatially blocked CV**



Due to the small sample size, fifteen different 80/20 train-test splits were generated

## CLASSICAL MACHINE LEARNING VERSUS DEEP LEARNING FOR DATA-SCARCE CROPLAND MAPPING: A CASE STUDY OF IRAN

using `GroupShuffleSplit` in a second, higher-power comparison that also respected the same geographical groups (Table 4.2).

| Model         | Kappa         |
|---------------|---------------|
| Random Forest | 0.913 ± 0.030 |
| XGBoost       | 0.910 ± 0.046 |
| CatBoost      | 0.919 ± 0.022 |
| TempCNN       | 0.899 ± 0.052 |

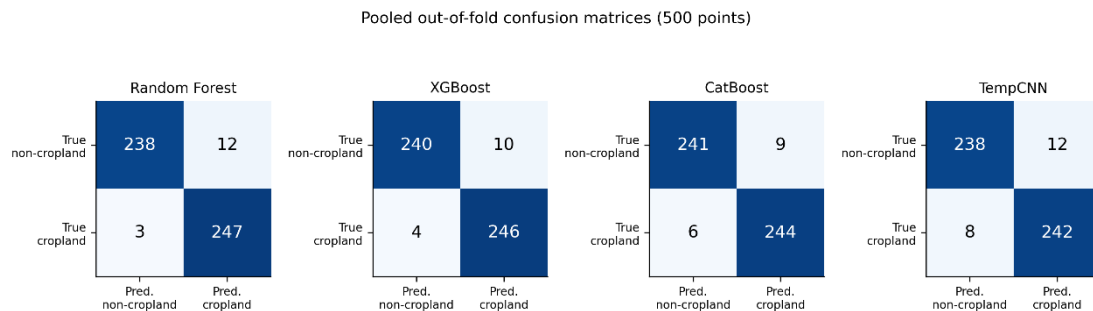
**Table 4.2: Repeated-CV Kappa (15 GroupShuffleSplit splits)**

For both comparisons, paired significance testing (a paired t-test and the Wilcoxon signed-rank test) was used. The paired t-test p-values for TempCNN's Kappa deficit against Random Forest, XGBoost, and CatBoost throughout the five-fold comparison were 0.33, 0.30, and 0.19 (Wilcoxon: 0.63, 0.19, 0.19). The identical three p-values for the fifteen-split comparison were 0.16, 0.28, and 0.12 (Wilcoxon: 0.22, 0.42, 0.23). Although the p-values from the higher-power comparison are consistently tighter than those from the five-fold comparison, none of them meet the traditional  $p < 0.05$  criteria.

This ranking is consistent with pooled out-of-fold confusion matrices, which are calculated by combining all five cross-validation folds' held-out predictions for each model into a single 500-point confusion matrix (Table 4.3).

| Model         | Accuracy | TN  | FP | FN | TP  |
|---------------|----------|-----|----|----|-----|
| Random Forest | 0.97     | 238 | 12 | 3  | 247 |
| XGBoost       | 0.97     | 240 | 10 | 4  | 246 |
| CatBoost      | 0.97     | 241 | 9  | 6  | 244 |
| TempCNN       | 0.96     | 238 | 12 | 8  | 242 |

**Table 4.3: Pooled out-of-fold confusion matrices and accuracy**



### 4.5.2 Feature importance

Each of the three traditional models was fitted once on the entire training set with adjusted hyperparameters in order to extract native feature importance. A limited set of Landsat-8-derived vegetation and moisture indices (NDMI, NDVI, and NDWI) from a single month, March 2020, the ninth month of the twelve-month feature window, dominate both Random Forest and XGBoost. There isn't a single dominant month in CatBoost's relevance ranking, which is much more dispersed between sensors and

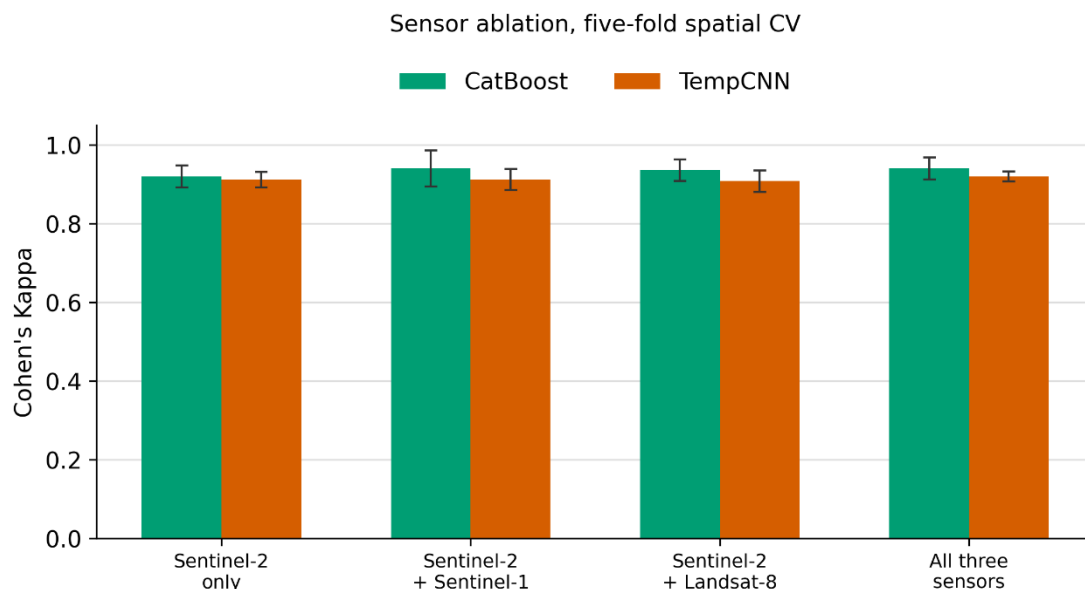
months.

#### 4.5.3 Sensor ablation

The cross-validated Kappa for CatBoost and TempCNN across four sensor configurations is shown in Table 4.4. Each model's pre-tuned hyperparameters are kept constant, with the only difference being which feature columns are included.

| Configuration           | CatBoost          | TempCNN           |
|-------------------------|-------------------|-------------------|
| Sentinel-2 only         | $0.920 \pm 0.028$ | $0.912 \pm 0.020$ |
| Sentinel-2 + Sentinel-1 | $0.940 \pm 0.046$ | $0.912 \pm 0.027$ |
| Sentinel-2 + Landsat-8  | $0.936 \pm 0.027$ | $0.908 \pm 0.027$ |
| All three sensors       | $0.940 \pm 0.028$ | $0.920 \pm 0.013$ |

**Table 4.4: Sensor ablation (Kappa, five-fold spatial CV)**



The majority of the discriminative signal is already captured by Sentinel-2 alone; Sentinel-1 and Landsat-8 each offer a comparable, modest secondary contribution on their own, but adding both together yields no additional gain over adding either one alone, suggesting that the two are significantly redundant with one another for this particular binary task. In all evaluated sensor configurations, not only the combined one, CatBoost's Kappa outperforms TempCNN's.

#### 4.5.4 Discriminating spectrally similar surfaces

The ground truth of the GeoAI Challenge is binary (cropland versus non-cropland), and Objective 5's spectrally-similar-classes query cannot be explicitly tested against crop-type sub-classes. ESA WorldCover (Zanaga et al., 2022), a global land-cover product with eleven distinct classes at 10-meter resolution, was sampled at each of the 500 Iran training points to address this issue. The pooled out-of-fold classification errors of each model were then cross-tabulated against the resulting class (Table 4.5).

# CLASSICAL MACHINE LEARNING VERSUS DEEP LEARNING FOR DATA-SCARCE CROPLAND MAPPING: A CASE STUDY OF IRAN

| WorldCover class       | Points | CatBoost errors | TempCNN errors |
|------------------------|--------|-----------------|----------------|
| Cropland               | 306    | 13              | 16             |
| Tree cover             | 75     | 2               | 1              |
| Grassland              | 65     | 0               | 0              |
| Shrubland              | 31     | 0               | 0              |
| Bare/sparse vegetation | 10     | 0               | 0              |
| Built-up               | 13     | 0               | 0              |

**Table 4.5: Out-of-fold errors by WorldCover land-cover class**

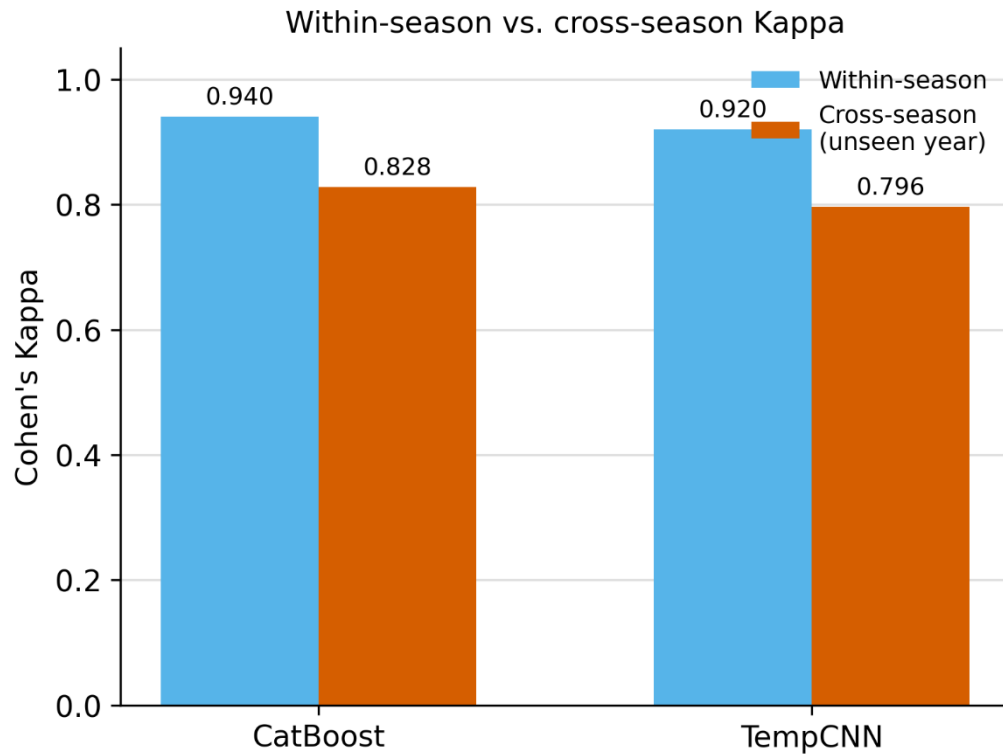
Grassland, shrubland, bare or scant vegetation, and built-up areas are all accurately classified by both models. Nearly all of the errors made by either model (87% for CatBoost and 94% for TempCNN) occur in areas that WorldCover independently classifies as cropland. This means that the errors made by both models focus on surfaces that are actually agricultural rather than being mistaken for clearly different types of land cover.

## 4.5.5 Temporal extendibility

Evaluating both models against a time period different from training was necessary for Objective 6. With the premise that each point's farmland status remained constant over the intervening year, each model was trained on the entire July 2019–June 2020 season and assessed against a second extracted year of characteristics (July 2020–June 2021) at the same 500 points using the same labels (Table 4.6).

| Model    | Within-season Kappa | Cross-season Kappa | Drop   |
|----------|---------------------|--------------------|--------|
| CatBoost | 0.940               | 0.828              | −0.112 |
| TempCNN  | 0.920               | 0.796              | −0.124 |

**Table 4.6: Cross-season Kappa (trained Jul2019-Jun2020, evaluated Jul2020-Jun2021)**



Both models lose performance against the unseen year; CatBoost's cross-season Kappa remains higher than TempCNN's, and by a wider absolute margin (0.032) than the within-season gap between the two (0.020).

#### 4.6 Summary

The data understanding, preparation, modelling, and evaluation work completed under the CRISP-DM methodology outlined in Chapter 3 has been presented in this chapter, along with a detailed description of a real evaluation error found between two modelling iterations and its implications for each downstream result. In all comparison runs, including within-season cross-validation (both standard and higher-power repeated), a four-configuration sensor ablation, and a temporal extendibility test, the performance of the tuned deep learning model is matched or surpassed by classical gradient boosting, spearheaded by CatBoost, where the difference grows rather than shrinks. Neither model demonstrates a significant advantage in differentiating between obviously distinct land cover and cropland; instead, the mistakes of both models focus nearly exclusively on surfaces that are indeed cropland-like. These findings are now critically discussed in Chapter 5 in relation to the goals of this dissertation as well as the literature studied in Chapter 2.

## CHAPTER 5: DISCUSSION

### *5.1 Introduction*

This chapter examines the findings in Chapter 4 in light of the goals and objectives of this dissertation (Chapter 1) as well as the body of current research reviewed in Chapter 2. The results of this dissertation are compared to the nearest published study in Section 5.2. The results themselves are rigorously assessed in Section 5.3, along with the methodological error covered in Chapter 4 and the statistical confidence they support. Each of the seven goals of this dissertation is critically assessed in turn in Section 5.4. The project's overall constraints are discussed in Section 5.5. The scalability of the assessed pipeline to Sudan and Afghanistan is covered in Section 5.6 (Objective 7).

### *5.2 Comparison with Existing Research*

The main conclusion of this dissertation, that under the training constraint of the GeoAI Challenge, classical gradient boosting matches or slightly outperforms a tailored temporal convolutional network's performance, fits well within rather than outside of the literature covered in Chapter 2. It is in line with the actual results of the initial 2023 GeoAI Challenge, when deep learning was not applied in any of the publicly available top-placing solutions, but rather classical gradient boosting (Section 2.4). Although their Temporal CNN remained competitive rather than obviously the weakest of their four models compared, it is consistent with the closest available published comparator, Mohammadi Mobarakeh, Pourmanafi, and Ahmadi's (2025) arid-Iran study, whose own strongest performer was Random Forest rather than their Temporal CNN. This dissertation's own results (Table 4.1) also broadly reproduce this pattern. Additionally, it is generally in accordance with the Nigerian study by Gajardo et al. (2025), which found that an existing classical baseline eventually outperformed a specially designed deep learning model that was trained on relatively little local data (Section 2.7).

The main difference between the results of this dissertation and some of the literature, particularly Pelletier, Webb, and Petitjean's (2019) original result and Ji et al.'s (2018) rice-corn case (Section 2.5), is that the deep learning advantage of those studies was shown on datasets that were three to four orders of magnitude larger than the 500 samples available here. The evidence presented in this dissertation is consistent with the interpretation that, rather than no such advantage exists in principle, TempCNN's architectural advantages require more training data to realise than the GeoAI Challenge's constraint permits. This interpretation was already predicted by Section 2.5's review of the literature prior to the existence of this dissertation's own results.

According to the modified spectrally-similar-surfaces analysis (Section 4.5.4), both model families make the vast majority of their mistakes on surfaces that are actually cropland-like rather than confusing cropland with distinctly different land cover, and there is no discernible difference in this pattern between the deep learning and classical approaches.

### *5.3 Critical Evaluation of the Results*

### 5.3.1 The methodological correction and what it demonstrates

The modelling inaccuracy detailed in Section 4.4.3-4.4.4 should not be left out of a critical assessment of this dissertation's findings because it significantly altered the main conclusion. For three reasons, this is viewed here as proof of methodological practice that was successful rather than unsuccessful. First, the problem was discovered internally, before to submission, by actively examining an exceptionally strong result instead of passively accepting it; an exceptionally good result is precisely the kind of finding that should lead to more thorough investigation, not less. Second, compared to the uncorrected result (Iteration 1), the corrected result (Iteration 2) is more consistent with the rest of the evidence gathered in this dissertation. It is consistent with the pattern found throughout the literature reviewed in Section 5.2 and with the temporal extendibility finding (Section 4.5.5), where CatBoost already outperformed TempCNN even under the Iteration 1 evaluation. Third, compared to the uncorrected result, the corrected result is also more in line with the nearest published comparative benchmark (Mohammadi Mobarakeh, Pourmanafi, and Ahmadi, 2025). A result that had to be walked back after publication would have been a true failure of rigour; on the other hand, a result that was discovered, fixed, and then determined to be more consistent with independent evidence is proof that the assessment approach worked as intended overall.

### 5.3.2 How much confidence the evidence supports

In comparison to the variability between cross-validation folds, the changes presented in Chapter 4 are small in absolute size, despite being constant in direction across all comparison runs. At the traditional  $p < 0.05$  level, neither the five-fold nor the higher-power fifteen-split comparison revealed a statistically significant difference between TempCNN and a classical model (Section 4.5.1). The lack of statistical significance at this sample size does not imply that there is no difference, so this should be read cautiously rather than as a null result. This study is underpowered to detect a Kappa difference of the size observed (0.011-0.024 across the various comparisons run) with high confidence because only 500 labelled points are available under the GeoAI Challenge's constraint. This is a direct and expected consequence of the training-sample cap this dissertation purposefully operates under (Section 1.2) rather than a flaw in the evaluation design. The entire body of evidence, including within-season cross-validation, the higher-power repeated comparison, the sensor ablation, the temporal extendibility test, and the pooled confusion matrices, all point to the same conclusion: classical gradient boosting demonstrates a slight, consistent, but statistically unconfirmed advantage over TempCNN on this task. This dissertation makes a claim that is more measured and honestly earned than either "TempCNN is significantly worse" or "no difference exists between the methods."

## 5.4 Critical Evaluation of the Objectives

Here, each of the seven goals outlined in Section 1.2 is evaluated in relation to actual accomplishments. Objectives 1-4 (literature review, data acquisition and feature engineering, the classical baseline, and the deep learning model) were accomplished as intended. Two real technical flaws were discovered and fixed during the process (Sections 4.2.3, 4.3.2), and for Objective 4 in particular, a more significant evaluation flaw was discovered and fixed between two modelling iterations (Sections 4.4.3-4.4.4). This is arguably a stronger demonstration of rigorous execution than an entirely error-free process would have been. Because of the binary ground truth available, the spectrally-similar-\*classes\* question, which is framed around crop-type discrimination in the

literature this dissertation builds upon, had to be tested using an independent land-cover product as a substitute for genuine crop-type labels (Section 4.5.4). This was a reasonable and informative adaptation, but it was not a literal fulfilment of the objective as originally stated. This dissertation's most important single finding—that a model's within-season ranking does not always predict its cross-season ranking and that the dissertation's classical-over-deep-learning finding actually strengthens rather than weakens under this more practically demanding test—was produced by achieving Objective 6, temporal extendibility. Objective 7, the scalability discussion, is addressed in full in Section 5.6 below.

### 5.5 Limitations

Here, a number of the work's weaknesses are clearly acknowledged.

**Sample size and statistical power:** Although a purposeful and motivating aspect of the original challenge's design, the training-sample cap of 500 points restricts the statistical confidence that can be attained in each model comparison (Section 5.3.2).

**The binary-label limitation:** the WorldCover-based replacement for Objective 5 should not be overly interpreted as directly supporting or contradicting the crop-type-specific findings reported elsewhere in the literature. Instead, it tests discrimination between cropland and other broad land-cover classes, not between crop types. The ILULC-2022 dataset (Karimi et al., 2024), which was found to be a better fit to the original frame, could not be obtained in the time allotted for this dissertation; Chapter 6 revisits this as a specific direction for future work.

**The temporal extendibility stable-label assumption:** The cross-season evaluation is predicated on the temporal extendibility stable-label assumption that each point's cropland status did not change between the training year and the evaluation year. This assumption cannot be confirmed by the available data and is likely to be violated for at least some points in an area with the type of small, fragmented, rotated smallholder cultivation described in Section 2.4. A Support Vector Machine baseline that was included in the closest published comparative benchmark (Mohammadi Mobarakeh, Pourmanafi, and Ahmadi, 2025) was not incorporated due to a scope decision that prioritised the depth of evaluation technique over the range of models examined.

**Data quality:** Since consistent model disagreement is suggestive of potential label noise but is not, by itself, sufficient evidence to overrule the original ground truth, the eleven points that were independently misclassified by both model families (Section 4.3.4) were neither eliminated nor relabelled.

**A single region:** This chapter's empirical findings only apply to Iran; Section 5.6 speculates about how they can be applied to Sudan and Afghanistan without going into full detail.

### 5.6 Scalability to Sudan and Afghanistan

Without complete deployment and assessment, Objective 7 requires a critical discussion of how the pipeline and conclusions created for Iran would need to be modified to extend to the other two regions of the GeoAI Challenge. Using the target-period and regional facts already established for this dataset (Section 4.2.2) as well as the actual agricultural and climatic context of each region, the discussion that follows separates what would transfer with little to no change from what would require true, region-specific rework.

#### 5.6.1 What transfers directly

By construction, a significant portion of the pipeline is not unique to Iran and would continue with little modification. Since the data from all three regions are contained in the same files, the regional filtering rule that was used to separate Iran's 500 training points from the same Train.csv/Test.csv files (Section 4.2.2) also applies to Sudan and Afghanistan. Once the region's actual target period and any region-specific cloud-masking behaviour have been verified, retargeting the Earth Engine feature-extraction pipeline (Section 4.3) to a different region and period is a configuration change rather than a rewrite because it is already parameterised by date range and point set rather than hardcoded to Iran.

Although each reported number would need to be re-derived from scratch for each region's unique data, the model architectures, hyperparameter search methodology, spatially-blocked cross-validation approach, and statistical significance testing methodology (Sections 4.4–4.5) are all general-purpose and would apply unchanged in mechanism. Since WorldCover is a global product, the WorldCover-based replacement for spectrally-similar-surface analysis (Section 4.5.4) is also directly reusable; however, the specific land-cover classes present and, consequently, the specific confusions found would need to be established freshly for each region rather than assumed to match Iran's pattern.

#### 5.6.2 Sudan: a closer fit in period, a different agricultural system

The twelve-month feature-extraction design developed for Iran applies to Sudan without requiring a separate temporal framework, unlike Afghanistan, since Sudan shares Iran's exact target timeframe, July 2019 to June 2020 (Section 5.6.3). However, there are significant differences in the underlying agricultural system. Sorghum, millet, sesame, and groundnuts are the main crops of the traditional and semi-mechanized rainfed subsectors, and the majority of Sudan's cultivated land—roughly 95% by area—is rainfed rather than irrigated. The country's central clay plains and semi-arid zones experience a single, monomodal rainy season that runs from May to October (FAO, 2015).

This has two specific ramifications for a direct pipeline expansion of this dissertation. First, it is quite likely that the vegetation-index signal that this dissertation discovered concentrated in a single month for Iran (Section 4.5.2) will concentrate much more closely for Sudan, most likely during the May–October rains rather than being dispersed throughout the year. Second, cloud cover during this rainy window poses a substantial risk to the Sentinel-2 signal at the critical moment, a more serious form of the Landsat-8 revisit-gap issue that Iran has already experienced (Section 4.3.3).

In contrast to Iran's more clearly defined smallholder plots, Sudan's rainfed, erratic-yield cropping system is likely more difficult to distinguish spectrally from natural semi-arid vegetation. This would require direct testing using Sudan's own WorldCover-substitute analysis rather than assuming to replicate this dissertation's Iran-specific finding that errors cluster almost entirely within cropland-like surfaces.

### 5.6.3 Afghanistan: a genuinely different temporal design problem

Compared to Iran's year-long window, Afghanistan's target period, April 2022 in Nangarhar province specifically (Zindi, 2023), is both a different absolute period and fundamentally narrower; the competition's own starter notebook extracted imagery for April-May 2022 specifically, confirming this as roughly a two-month rather than a twelve-month target. The province of Nangarhar grows a lot of maize, but its agriculture is irrigated rather than rainfed, relying on the Kabul and Kunar rivers and their canal networks, which are themselves dependent on the timing of snowmelt in a hilly catchment (Akhtar et al., 2018).

The twelve-monthly-composite time-series design developed for Iran is not naturally supported by a single narrow target window, and expanding this dissertation's pipeline to Afghanistan would necessitate a genuine design decision rather than a configuration change: either creating a shorter within-season sequence (for example, extending backward from April across the preceding several months to capture early-season green-up ahead of the target date) or going back toward something closer to the single-composite approach used in the unmodified starter notebook, which this dissertation's own conclusion that a genuine temporal sequence materially benefits the deep learning model (Section 4.3.1).

Additionally, Sudan's rainfed system and Iran's mixed smallholder cropping are sufficiently different from snowmelt-driven, canal-irrigated agriculture over a limited spring window that no conclusions drawn from either of the other two regions—including the results of this dissertation for Iran—should be extrapolated to Afghanistan without direct testing.

### 5.6.4 Summary

The problem's basic architecture is the same in all three regions, with a hard cap on labelled training samples and a multi-sensor Sentinel-1/Sentinel-2/Landsat-8 feature basis. Every public top-placing solution to the initial challenge in all three regions—not just Iran—has already confirmed this as a reasonable general strategy (ITU-GeoAI-Challenge, 2023).

However, extending this dissertation's particular pipeline and conclusions outside of Iran is a legitimate research task rather than a mechanical rerun: Afghanistan's narrow, snowmelt-irrigated target window and Sudan's rainfed, mono-modal cropping system each present a materially different problem from the one this dissertation actually solved, and the classical-over-deep-learning finding reported in Chapter 4 should be treated as a hypothesis to re-test in each region rather than a result assumed to generalise automatically.

## 5.7 Summary

The results of this dissertation have been compared to the closest published comparative research in this chapter. The corrected finding, classical gradient boosting matching or surpassing a tuned deep learning model under a severe training-sample constraint, is consistent with both the closest available benchmark study and the actual result of the 2023 challenge.

It has been clear about the bounds of statistical confidence that the available sample size supports and has critically assessed the results themselves, treating the methodological correction discovered during review as an example of rigour rather than a weakness. Each of the seven goals outlined in Chapter 1 has been compared to what was really accomplished, and it has highlighted how the evaluated pipeline would need to be modified for Sudan and Afghanistan without going into full implementation. The

dissertation is now concluded in Chapter 6.

## 6. CONCLUSION

### 6.1 *Summary of the Dissertation*

The goal of this dissertation was outlined in Chapter 1: to develop, implement, and critically evaluate machine learning and deep learning methods for annual cropland extent mapping over Iran as a data-scarce, semi-arid case study region. The goal was to determine whether deep learning offers quantifiable advantages over an established gradient-boosting baseline under a severely constrained training budget, and to evaluate how the resulting pipeline could be scaled to Sudan and Afghanistan. Each of the seven objectives set out in Section 1.2 has been addressed and critically evaluated in Chapter 5 (Section 5.4); in brief, the literature review (Objective 1) identified three specific gaps this dissertation addresses, the data pipeline and both model families were implemented as planned (Objectives 2-4), comparative evaluation was carried out in an adapted form necessitated by the dataset's binary labels (Objective 5), temporal extendibility was evaluated and produced this dissertation's most decisive finding (Objective 6), and the scalability of the pipeline to Sudan and Afghanistan was critically discussed without full implementation (Objective 7).

The main, convincing conclusion of this dissertation is that, under the 500-point training constraint of the GeoAI Challenge, classical gradient boosting, best represented by CatBoost and closely matched by Random Forest and XGBoost, meets or slightly surpasses the performance of a temporal convolutional deep learning model on binary cropland classification for Iran, both within the training season (Kappa 0.940 versus 0.920) and, with a larger margin, when generalising to an unseen year (Kappa 0.828 versus 0.796). As a direct and expected result of the training-sample cap this dissertation purposefully operates within, this advantage is directionally consistent across all analysis runs in this dissertation but does not reach conventional statistical significance at the sample sizes available (Section 5.3.2). The identification and correction of a genuine evaluation error that had initially led to the opposite headline conclusion—described in detail in Sections 4.4 and 5.3.1—is a methodological finding worth restating independently from the substantive results. The process of identifying an exceptionally strong result as a signal to investigate rather than a result to accept and tracing it to its actual cause is itself a demonstration of the methodological rigour this dissertation set out to practise.

### 6.2 *Research Contributions*

Three distinct additions are made by this dissertation, each of which fills a gap noted in Chapter 2 (Section 2.10). The first reported deep learning baseline for the 2023 GeoAI Challenge for Cropland Mapping dataset is implemented and evaluated; no publicly available solution to the original challenge tried this. Second, under the training-sample constraint of this challenge, it offers a directly comparable, methodologically rigorous test of the classical-versus-deep-learning question for cropland classification. This test includes a higher-power repeated significance test that is rarely used in the closely related literature reviewed in Chapter 2, the majority of which reports single point-estimate comparisons without any formal significance testing. Third, it assesses temporal extendibility, the second goal of the original challenge, a dimension that was largely ignored by both the challenge's public solutions and the majority of the broader literature reviewed in Section 2.8. It discovers that a model's within-season ranking does not always

predict its cross-season ranking, an observation that is directly applicable to any future deployment of a cropland-mapping model for real operational, as opposed to one-time academic, use.

Practically speaking, these results imply that a future implementer should not assume that the more architecturally complex option is safer or more accurate when deciding between a temporal deep learning model and classical gradient boosting for a similarly data-constrained cropland mapping task; under the particular conditions tested here, it was neither.

### *6.3 Limitations and Future Research and Development*

The main limitations of this dissertation are covered in detail in Section 5.5 and are briefly summarised here: statistical power is limited by the 500-point training constraint; the dissertation's original crop-type-level hypothesis could not be directly tested due to the binary ground truth; the temporal extendibility test is based on an unverifiable stable-label assumption; a Support Vector Machine baseline, which was present in the closest published comparator, was not implemented; and eleven training points were identified as potential label-noise candidates but were left unresolved. The main conclusion of this dissertation is not invalidated by any of these constraints, but they do restrict the degree of confidence with which it should be applied outside of the particular settings examined. Each of these limits also offers a distinct path for further research.

**Complete execution for Afghanistan and Sudan.** Retargeting the feature-extraction pipeline to each region's actual conditions and directly testing whether this dissertation's classical-over-deep-learning finding for Iran holds, weakens, or reverses under each region's unique agricultural and climatic conditions is the most direct extension of this dissertation. This implementation is only covered in Section 5.6.

**Resolving the crop-type-label limitation.** The spectrally-similar-surfaces analysis could be repeated using its true crop-type sub-classes instead of the coarser WorldCover substitute if the ILULC-2022 dataset (Karimi et al., 2024), which was found but not accessed during this dissertation's timeframe, becomes available. This would test the main hypothesis of this dissertation in the manner that the reviewed literature initially frames it.

**A larger or actively expanding training set.** The 500-point limitation, not the models or methodology employed, is the direct and accepted cause of this dissertation's inability to demonstrate statistical significance. Instead of merely asking for a larger labelling budget, Ji et al.'s (2018) demonstrated active-learning efficiency gain, achieving a given accuracy with about a quarter of the randomly-sampled data, is a tangible, proven way to extract more statistical power from an equally constrained labelling budget.

**Manual review of candidate label-noise points.** In order to ascertain whether the eleven points identified in Section 4.3.4 as consistently misclassified by both model families represent actual label errors, genuinely ambiguous or mixed-use parcels, or simply cases that both models happen to find equally difficult for unrelated reasons, they could be manually examined against the original satellite imagery and, where genuinely available, higher-resolution reference imagery.

**A repeated multi-year temporal extendibility test.** The temporal extendibility test for this dissertation contrasts precisely two years. The stable-label assumption could be indirectly verified by extending this to three or more consecutive years, if satellite archive

availability allows. This is because a truly stable-label point should exhibit consistent classification throughout all tested years, whereas a point whose true land use changed should exhibit a discernible, dateable shift.

#### *6.4 Personal Reflections*

One of the most difficult and beneficial aspects of my MSc has been this dissertation since it forced me to think more critically about how research findings are generated and verified rather than just applying machine learning techniques. When I started the study, my focus was on the technical challenge of whether a temporal deep learning model could perform better than well-established traditional machine learning techniques with limited labelled data. I had a far more comprehensive grasp of the significance of experimental design, data quality, validation, and critical interpretation by the project's conclusion.

Finding a mistake in my own evaluation process taught me one of the most crucial lessons. The deep learning model had significantly outperformed all three of the conventional models, according to the initial TempCNN findings, which were exceptionally impressive. I saw that several of the downstream findings did not mathematically agree with the quoted performance, therefore I decided not to accept this result right now. After looking into this disparity, I discovered that the assessment fold's use during epoch selection was flawed. I made the necessary corrections and repeated the pertinent modelling and evaluation process, which significantly altered the outcome and flipped the models' apparent ranking. This meant that work had to be repeated, which was initially annoying, but in the end, it turned out to be one of the most beneficial aspects of the project. It taught me that an exceptionally positive outcome should not always be interpreted as proof of success but rather as a cause for additional research. It also revealed a flaw in my first validation strategy, which concentrated on assessing the models' performance without putting in place robust enough checks that would have spotted issues with the review process sooner. In subsequent projects, I would incorporate independent validation tests from the start of the modelling process, especially those that compare various evaluation paths and highlight outcomes that don't seem to match the underlying predictions.

My perception of what makes a successful research study was also altered by the experience. At first, I thought a strong dissertation would be one in which the suggested deep learning method clearly outperformed the classical baseline. Rather, in the studied conditions, the corrected results demonstrated that the traditional techniques consistently matched or marginally exceeded TempCNN. I discovered that the goal of study is to determine what the existing evidence can support rather than to validate a favoured hypothesis, even when this did not support the outcome I had first anticipated. The significance of upholding scientific objectivity became more evident to me because the result deviated from the original expectation.

My perspective on data scarcity was also altered by working with a purposefully small training dataset. Prior to this study, I primarily thought of a tiny dataset as a model training challenge. Through the dissertation, I came to understand that model performance is not the only factor affected by a lack of labelled data. Additionally, it limits statistical power, increases the uncertainty of model comparisons, and affects the degree of confidence with which conclusions can be generalised. This was especially noticeable when the model differences were consistent in direction but fell short of traditional statistical significance. As a result, I started to be more careful about using minor numerical variations as conclusive proof that one approach was better than another.

## **CLASSICAL MACHINE LEARNING VERSUS DEEP LEARNING FOR DATA-SCARCE CROPLAND MAPPING: A CASE STUDY OF IRAN**

My capacity to work independently with new data and technologies was also enhanced by the project. Building a multi-sensor feature-extraction pipeline, working with satellite imagery via Google Earth Engine, implementing and fine-tuning multiple machine learning techniques, and investigating computational and data-quality issues as they emerged were all tasks that required me to look into the provided challenge material rather than assuming it was accurate. Finding issues with the initial beginner material emphasised how crucial it is to comprehend what code is actually doing rather than assuming that an existing implementation is always dependable.

Overall, I think the dissertation has helped me become a better researcher and data scientist. I now feel more at ease challenging unexpected outcomes, pointing out flaws in my own methods, repeating work as needed, and differentiating between what the evidence shows and what I had anticipated. I anticipate applying these insights to upcoming data science initiatives, especially those including machine learning experiments, where seemingly minor methodological choices can have a significant impact on the conclusions reached.

## REFERENCES

- Akhtar, F., Awan, U.K., Tischbein, B. and Liaqat, U.W. (2018) 'Assessment of Irrigation Performance in Large River Basins under Data Scarce Environment—A Case of Kabul River Basin, Afghanistan', *Remote Sensing*, 10(6), 972. Available at: <https://doi.org/10.3390/rs10060972>
- Alami Machichi, M., El Mansouri, L., Imani, Y., Bourja, O., Lahlou, O., Zennayi, Y., Bourzeix, F., Hanadé Houmma, I. and Hadria, R. (2023) 'Crop mapping using supervised machine learning and deep learning: a systematic literature review', *International Journal of Remote Sensing*, 44(8), pp. 2717–2753. Available at: <https://doi.org/10.1080/01431161.2023.2205984>
- Chen, T. and Guestrin, C. (2016) 'XGBoost: A Scalable Tree Boosting System', in *Proceedings of the 22nd ACM SIGKDD International Conference on Knowledge Discovery and Data Mining*, pp. 785–794. Available at: <https://doi.org/10.1145/2939672.2939785>
- Food and Agriculture Organization of the United Nations (FAO) (2023) 'Participate now in the GEO-AI Challenge for Cropland Mapping'. Available at: <https://www.fao.org/agroinformatics/news/news-detail/geo-ai-challenge-for-cropland-mapping-by-itu/en> (Accessed: 7 July 2026).
- Food and Agriculture Organization of the United Nations (FAO) (2015) *National Investment Profile: Water for Agriculture and Energy — Sudan*. Khartoum/Rome: FAO. Available at: [https://www.fao.org/fileadmin/user\\_upload/agwa/docs/NIP\\_Sudan\\_Final.pdf](https://www.fao.org/fileadmin/user_upload/agwa/docs/NIP_Sudan_Final.pdf)
- Gajardo, J., Volpi, M., Onwude, D. and Defraeye, T. (2025) 'Evaluating the role of training data origin for country-scale cropland mapping in data-scarce regions: A case study of Nigeria', *ISPRS Open Journal of Photogrammetry and Remote Sensing*. Available at: <https://arxiv.org/abs/2312.10872>
- Gorelick, N., Hancher, M., Dixon, M., Ilyushchenko, S., Thau, D. and Moore, R. (2017) 'Google Earth Engine: Planetary-scale geospatial analysis for everyone', *Remote Sensing of Environment*, 202, pp. 18–27. Available at: <https://doi.org/10.1016/j.rse.2017.06.031>
- ITU-GeoAI-Challenge (2023) '6th-Place-Cropland-Mapping'. Available at: <https://github.com/ITU-GeoAI-Challenge/6th-Place-Cropland-Mapping> (Accessed: 7 July 2026).
- Ji, S., Zhang, C., Xu, A., Shi, Y. and Duan, Y. (2018) '3D Convolutional Neural Networks for Crop Classification with Multi-Temporal Remote Sensing Images', *Remote Sensing*, 10(1), 75. Available at: <https://doi.org/10.3390/rs10010075>
- Karimi, N., Sheshangosht, S., Rashtbari, M., Torabi, O., Sarbazvatan, A., Lari, M., Aminzadeh, H., Abolhoseini, S. and Eftekhari, M. (2024) 'An advanced high resolution land use/land cover dataset for Iran (ILULC-2022) by focusing on agricultural areas based on remote sensing data', *Computers and Electronics in Agriculture*, 228, 109677. Available at: <https://doi.org/10.1016/j.compag.2024.109677>

**CLASSICAL MACHINE LEARNING VERSUS DEEP LEARNING FOR DATA-SCARCE CROPLAND  
MAPPING: A CASE STUDY OF IRAN**

Mahesh, P. and Soundrapandiyan, R. (2024) 'Yield prediction for crops by gradient-based algorithms', *PLOS ONE*, 19(8), e0291928. Available at: <https://doi.org/10.1371/journal.pone.0291928>

Metzger, N., Turkoglu, M.O., D'Aronco, S., Wegner, J.D. and Schindler, K. (2022) 'Crop Classification Under Varying Cloud Cover With Neural Ordinary Differential Equations', *IEEE Transactions on Geoscience and Remote Sensing*, 60, pp. 1–12. Available at: <https://doi.org/10.1109/TGRS.2021.3101965>

Mohammadi Mobarakeh, Z., Pourmanafi, S. and Ahmadi, M. (2025) 'Employing Sentinel-2 time-series and noisy data quality control to enhance crop classification in arid environments: A comparison of machine learning and deep learning methods', *International Journal of Applied Earth Observation and Geoinformation*, 142, 104678. Available at: <https://doi.org/10.1016/j.jag.2025.104678>

Moumni, A. and Lahrouni, A. (2021) 'Machine Learning-Based Classification for Crop-Type Mapping Using the Fusion of High-Resolution Satellite Imagery in a Semiarid Area', *Scientifica*, 2021, Article 8810279. Available at: <https://doi.org/10.1155/2021/8810279>

Paszke, A., Gross, S., Massa, F., Lerer, A., Bradbury, J., Chanan, G., Killeen, T., Lin, Z., Gimelshein, N., Antiga, L., Desmaison, A., Kopf, A., Yang, E., DeVito, Z., Raison, M., Tejani, A., Chilamkurthy, S., Steiner, B., Fang, L., Bai, J. and Chintala, S. (2019) 'PyTorch: An Imperative Style, High-Performance Deep Learning Library', *Advances in Neural Information Processing Systems*, 32.

Pedregosa, F., Varoquaux, G., Gramfort, A., Michel, V., Thirion, B., Grisel, O., Blondel, M., Prettenhofer, P., Weiss, R., Dubourg, V., Vanderplas, J., Passos, A., Cournapeau, D., Brucher, M., Perrot, M. and Duchesnay, É. (2011) 'Scikit-learn: Machine Learning in Python', *Journal of Machine Learning Research*, 12, pp. 2825–2830.

Pelletier, C., Webb, G.I. and Petitjean, F. (2019) 'Temporal Convolutional Neural Network for the Classification of Satellite Image Time Series', *Remote Sensing*, 11(5), 523. Available at: <https://doi.org/10.3390/rs11050523>

Prokhorenkova, L., Gusev, G., Vorobev, A., Dorogush, A.V. and Gulin, A. (2018) 'CatBoost: unbiased boosting with categorical features', *Advances in Neural Information Processing Systems*, 31.

Reuss, J., Macdonald, J., Becker, S., Richter, L. and Körner, M. (2025) 'The EuroCropsML time series benchmark dataset for few-shot crop type classification in Europe', *Scientific Data*, 12(1), 664. Available at: <https://doi.org/10.1038/s41597-025-04952-7>

Roberts, D.R., Bahn, V., Ciuti, S., Boyce, M.S., Elith, J., Guillera-Arroita, G., Hauenstein, S., Lahoz-Monfort, J.J., Schröder, B., Thuiller, W., Warton, D.I., Wintle, B.A., Hartig, F. and Dormann, C.F. (2017) 'Cross-validation strategies for data with temporal, spatial, hierarchical, or phylogenetic structure', *Ecography*, 40(8), pp. 913–929. Available at: <https://doi.org/10.1111/ecog.02881>

Rußwurm, M. and Körner, M. (2018) 'Multi-Temporal Land Cover Classification with Sequential Recurrent Encoders', *ISPRS International Journal of Geo-Information*, 7(4), 129. Available at: <https://doi.org/10.3390/ijgi7040129>

**CLASSICAL MACHINE LEARNING VERSUS DEEP LEARNING FOR DATA-SCARCE CROPLAND  
MAPPING: A CASE STUDY OF IRAN**

Sainte Fare Garnot, V. and Landrieu, L. (2020) 'Lightweight Temporal Self-Attention for Classifying Satellite Image Time Series', presented at the Workshop on Advanced Analytics and Learning on Temporal Data (AALTD, ECML-PKDD). Available at: <https://arxiv.org/abs/2007.00586>

Sainte Fare Garnot, V. and Landrieu, L. (2021) 'Panoptic segmentation of satellite image time series with convolutional temporal attention networks', in *Proceedings of the IEEE/CVF International Conference on Computer Vision (ICCV)*, pp. 4872–4881. Available at: [https://openaccess.thecvf.com/content/ICCV2021/html/Garnot\\_Panoptic\\_Segmentation\\_of\\_Satellite\\_Image\\_Time\\_Series\\_With\\_Convolutional\\_Temporal\\_ICCV\\_2021\\_paper.html](https://openaccess.thecvf.com/content/ICCV2021/html/Garnot_Panoptic_Segmentation_of_Satellite_Image_Time_Series_With_Convolutional_Temporal_ICCV_2021_paper.html)

Singh, G., Singh, S., Sethi, G. and Sood, V. (2022) 'Deep Learning in the Mapping of Agricultural Land Use Using Sentinel-2 Satellite Data', *Geographies*, 2(4), pp. 691–700. Available at: <https://doi.org/10.3390/geographies2040042>

Zanaga, D., Van De Kerchove, R., De Keersmaecker, W., Souverijns, N., Brockmann, C., Quast, R., Wevers, J., Grosu, A., Paccini, A., Vergnaud, S., Cartus, O., Santoro, M., Fritz, S., Georgieva, I., Lesiv, M., Carter, S., Herold, M., Li, L., Tsendbazar, N.E., Ramoino, F. and Arino, O. (2022) 'ESA WorldCover 10 m 2021 v200'. Available at: <https://doi.org/10.5281/zenodo.7254221>

Zhang, W., Guo, S., Zhang, P., Xia, Z., Zhang, X., Lin, C., Tang, P., Fang, H. and Du, P. (2023) 'A Novel Knowledge-Driven Automated Solution for High-Resolution Cropland Extraction by Cross-Scale Sample Transfer', *IEEE Transactions on Geoscience and Remote Sensing*, 61, pp. 1–16. Available at: <https://ieeexplore.ieee.org/document/10197441/>

Zhu, X.X., Tuia, D., Mou, L., Xia, G.S., Zhang, L., Xu, F. and Fraundorfer, F. (2017) 'Deep Learning in Remote Sensing: A Comprehensive Review and List of Resources', *IEEE Geoscience and Remote Sensing Magazine*, 5(4), pp. 8–36. Available at: <https://doi.org/10.1109/MGRS.2017.2762307>

Zindi (2023) 'GeoAI Challenge for Cropland Mapping with Satellite Imagery'. Available at: <https://zindi.africa/competitions/geo-ai-challenge-for-cropland-mapping-with-satellite-imagery> (Accessed: 7 July 2026).

## **APPENDIX A: ETHICAL APPROVAL**

### **A.1 Statement on Human participants**

This research involved no human participants at any stage. The dataset used throughout this dissertation consists of remotely sensed satellite imagery (Sentinel-1, Sentinel-2, Landsat8 and ESA WorldCover land-cover product acquired via Google Earth Engine free publicly available Earth observation archives and previously collected, anonymised geographic point coordinates and binary cropland/non-cropland labels, provided as part of the 2023 ITU/FAO/UNODC GeoAI Challenge for Cropland Mapping dataset, hosted on Zindi and licensed CC-BY-SA 4.0 for research and educational purposes.

No survey, interview, questionnaire or any form of interaction with human subjects took place. No personally identifiable information was collected, processed or stored at any point in this project.

### **A.2 Basis for not requiring formal ethical approval**

On the basis of Section A.1, this project doesn't fall within the categories of research that typically require a full ethical review.

### **A.3 Data licensing and access**

All data used in this dissertation is freely and openly available under licenses that explicitly permit research and educational use, satisfying this dissertation's reproducibility requirements.

- The GeoAI Challenge dataset (Zindi, 2023) is licensed CC-BY-SA 4.0
- Sentinel-1 and Sentinel-2 imagery and Landsat-8 are freely available under their respective open data policies
- ESA WorldCover (Zanaga et al., 2022) is freely available under CC-BY 4.0 license

## APPENDIX B: OTHER APPENDICES

### B.1 Requirement Analysis

#### B.1.1 Introduction

This section provides a requirement analysis that supports the design choices outlined in Chapters 3-4, differentiating between requirements that this dissertation independently adopted as an academic study rather than a competition re-entry and requirements that were directly inherited from the GeoAI Challenge's own dataset and rules. A traceability summary (Section B.1.5) maps each of the three categories of requirements—functional requirements (what the pipeline must do), non-functional requirements (the qualities it must have while doing so), and constraints (fixed conditions the project must operate within, not subject to design choice).

#### B.1.2 Functional Requirements

**FR1 – Regional and temporal scoping:** Instead of extracting satellite characteristics for a generic or default window, the pipeline must isolate the labelled points of a single region (Iran) from the shared, multi-region dataset. *Origin: derived from the challenge's own region/period structure (Section 4.2.2); this dissertation chose to scope to one region instead of all three (Section 1.1).*

**FR2 - Multi-sensor feature extraction:** Sentinel-1, Sentinel-2, and Landsat-8 imagery must be acquired and combined by the pipeline in accordance with the pattern found in all publicly available top-placing solutions to the initial challenge. *Origin: inherited, as shown in Section 2.6.*

**FR3 – Feature engineering beyond raw bands:** In accordance with the traditional literature discussed in Section 2.4, the pipeline must extract vegetation, water, and moisture indices from the raw spectral bands rather than depending just on raw reflectance data. *Origin: inherited practice; the choice of which particular indices to compute in this dissertation (Section 4.3.2).*

**FR4 – Temporal structure:** Since a temporal deep learning model needs a sequence to train from and classical models benefit from the same richer feature set, extracted features must represent a true time series spanning the goal period (monthly composites, Section 4.3.1) rather than a single static composite. *Origin: a necessity unique to this dissertation, not found in the unaltered beginning notebook (Section 4.2.3).*

**FR5 – Classical baseline models:** As a benchmark for the deep learning model, the pipeline must have at least one customised classical gradient-boosting model rather than one that is run with default hyperparameters. *Origin: derived from the predominant pattern among public solutions (Section 2.4); this dissertation made the particular choice to adjust all three and include Random Forest and XGBoost in addition to CatBoost (Section 4.4.1).*

**FR6 - Deep learning model:** Since there isn't a publicly available solution to the original problem, the pipeline needs to have a deep learning model that can train directly from the multi-temporal feature sequence. This dissertation fills this specific gap (Section 2.10). *Origin: the prerequisite of this dissertation.*

**FR7 – Fair, leakage-free evaluation:** The pipeline must employ a cross-validation

technique suitable for the geographical structure of the data and evaluate each model using a methodology that prevents information from the data being scored from influencing which model, configuration, or training checkpoint is reported. *Origin: general good practice in machine learning evaluation; the particular requirements for spatial-blocking and epoch-selection were directly sharpened by two problems that arose during the work of this dissertation (Sections 4.4.3 and 4.4.4) and are the most obvious example of a requirement that was discovered during the project rather than predetermined.*

**FR8 – Statistical confidence reporting:** Given the limited sample size available, comparative results must be presented using a statistical confidence measure rather than as point estimates. *Origin: the need for this dissertation, which fills a void in a large portion of the closely similar literature examined in Chapter 2, which usually presents single-run comparisons without formal significance testing.*

**FR9 – Temporal extendibility testing:** A trained model's evaluation against a time period different from its training data must be supported by the pipeline. *Origin: directly derived from the second stated purpose of the initial challenge (Section 4.5.5).*

### B.1.3 Non-Functional Requirements

**NFR1 – Reproducibility and open tooling:** Any future researcher who wants to duplicate this work must have unrestricted access to all software and data, without the need for proprietary or costly tools. *Origin: derived from the submission guidelines of the original challenge (which only allowed Python and Google Earth Engine) and strengthened by the academic integrity standards of this dissertation. satisfied using only open-source Python modules and the non-commercial use tier of Google Earth Engine (Section 3.4).*

**NFR2 – Licensing compliance:** Every piece of data utilised must have a license that permits its usage for research and teaching. *Origin: inherited; this use is specifically permitted by the GeoAI Challenge dataset's CC-BY-SA 4.0 license (see also Appendix A.3).*

**NFR3 – Computational feasibility within an MSc timeframe:** Within a reasonable dissertation deadline, the pipeline must be trainable and assessable on consumer-grade hardware without requiring specialist computational equipment. *Origin: this dissertation's own limitations, which directly influenced the choice to adopt TempCNN in particular among the deep learning architectures examined in Section 2.5, in part due to its relatively low architectural complexity, and to scope to one region in depth (Section 1.1) rather than three regions shallowly.*

**NFR4 – Methodological transparency:** Every design choice, including mistakes made and fixed, must be sufficiently detailed in the documentation for a third party to audit or replicate the work. *Origin: this dissertation's own requirement, fulfilled by the thorough implementation record that Chapters 3-5 refer to, and directly responsible for the discovery of the leaking bug in Section 4.4 as opposed to its silent reporting.*

**NFR5 - Data quality assurance:** Before being utilised for modelling, the engineering feature table must be examined for implausible values, duplication, and other fundamental quality flaws rather than being taken at face value. *Origin: this dissertation's own requirement, which is partially based on the noisy-data-quality-control phase outlined in the closest published comparator study (Mohammadi Mobarakeh, Pourmanafi, and Ahmadi, 2025) but does not replicate the particular methodology of that study (Section*

4.3.4).

#### B.1.4 Constraints

**C1 -Training-sample cap:** A fixed criterion of the original challenge's dataset design, a maximum of 1,000 labelled points per region (500 in this dissertation's Iran-only working subset) is available; this is not a parameter that this dissertation could modify. This restriction is addressed throughout this dissertation as a purposeful, practical aspect of the study subject rather than a barrier to be overcome. It is the immediate cause of the statistical-power limitation covered in Section 5.3.2.

**C2 – Fixed target period per region:** The target term for each region—Iran and Sudan: July 2019–June 2020; Afghanistan: April 2022—is set by the initial challenge and cannot be changed, which directly influences the viability and layout of the scalability debate in Section 5.6.

**C3 – No live leaderboard or ground truth for the held-out test set:** All of the quantitative results presented in Chapter 4 are inevitably generated from cross-validation on the labelled training subset rather than from a really held-out competition test set because the original competition is closed and the labels of 'Test.csv' are not accessible to this dissertation.

**C4 – Earth Engine quota tier:** The non-commercial Community usage tier of Earth Engine (150 EECU-hours/month) was used for feature extraction, which was adequate for the point-based, single-region extraction workload of this dissertation but a real ceiling that would require re-evaluation if the scalability discussion in Section 5.6 were fully implemented for two additional regions.

#### B.1.5 Traceability Summary

| Requirement        | Addressed in                                | Related objective (Section 1.2) |
|--------------------|---------------------------------------------|---------------------------------|
| FR1, FR9, C1, C2   | Section 4.2, Section 4.5.5, Section 5.6     | Objectives 2, 6, 7              |
| FR2, FR3, FR4      | Section 4.3                                 | Objective 2                     |
| FR5                | Section 4.4.1                               | Objective 3                     |
| FR6                | Section 4.4.2-4.4.4                         | Objective 4                     |
| FR7, FR8           | Section 4.4.3, Section 4.5.1, Section 5.3.2 | Objective 5                     |
| NFR1, NFR2, C3, C4 | Section 3.4                                 | (Underpins all objectives)      |
| NFR3               | Section 1.1, Section 4.4.2                  | Objective 4                     |
| NFR4               | Section 4.4.3-4.4.4, Section 5.3.1          | (Underpins all objectives)      |
| NFR5               | Section 4.3.4                               | Objective 2                     |